

An Application Development Project Report

BB84 Quantum Key Distribution Demonstrator For Secure Communication

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CERTIFICATE

This is to certify that this is the bonafide record of the application development entitled **"BB84 QKD"** submitted by **N.Spoorthi (2511CS040019), Saniya (2511CS040046), A.Akshitha (2511CS040015), M.Nikhitha (2511CS040006)** B. Tech II year I semester, Department of CSE (CS) during the year 2025-26. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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DECLARATION

We hereby declare that the project report entitled "**BB84 Quantum Key Distribution Demonstrator for Secure Communication**" has been carried out by us. This work has been submitted to the Department of Cyber Security, Malla Reddy University, Hyderabad in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology. we further declare that this project work has not been submitted in full or part for the award of any other degree in other educational institutions.

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ABSTRACT

The BB84 Quantum Key Distribution (QKD) Demonstrator for Secure Communication is a hardware-oriented project developed to demonstrate the basic principles of quantum-based key establishment. Conventional communication systems commonly rely on mathematical techniques to protect information, while quantum key distribution uses properties of quantum states to help two communicating parties establish a shared secret key and identify disturbances that may indicate eavesdropping.

The demonstrator follows the BB84 protocol, introduced by Charles Bennett and Gilles Brassard in 1984. In the protocol, a sender prepares bit values using randomly selected encoding bases and transmits corresponding quantum states to a receiver. The receiver independently chooses measurement bases. After transmission, both parties publicly compare basis choices, retain the positions where their bases match, and use a sample of the retained bits to estimate the error rate. If the observed error level is acceptable, the remaining data can be processed through classical post-processing to form a shared key; otherwise, the session is rejected.

The hardware implementation provides a tangible learning platform for understanding state preparation, transmission, measurement, basis reconciliation, and error checking. The exact optical or electronic components, communication medium, and measured performance should be documented according to the hardware actually assembled and tested. The project is intended as an educational demonstrator, not as a claim of production-grade quantum-secure communication.

Keywords: BB84, Quantum Key Distribution, Quantum Cryptography, Secure Communication, Quantum States, Basis Reconciliation, Eavesdropping Detection, Hardware Demonstrator.

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CHAPTER1

INTRODUCTION

The BB84 Quantum Key Distribution (QKD) Demonstrator for Secure Communication is a hardware-based project developed to demonstrate secure key exchange using the principles of quantum mechanics. In the modern digital world, protecting sensitive information from unauthorized access has become increasingly important. Traditional encryption techniques rely on mathematical algorithms to secure data, but advancements in computing technology have introduced new challenges to information security. Quantum Key Distribution offers a promising approach by enabling two communicating parties to establish a shared secret key and detect possible eavesdropping attempts during the communication process.

The BB84 protocol, introduced by Charles Bennett and Gilles Brassard in 1984, is one of the earliest quantum key distribution protocols. It allows a sender, known as Alice, and a receiver, known as Bob, to exchange key information using quantum states. Alice generates random bits and prepares corresponding quantum states using randomly selected bases. Bob measures the received states using randomly selected measurement bases. After transmission, both parties compare their basis selections through an authenticated classical communication channel and retain only the results obtained using matching bases. These results form a sifted key that can be processed further to establish a shared secret key.

The proposed project focuses on demonstrating the fundamental working principles of the BB84 protocol through a suitable hardware setup. The system represents the major stages of quantum key distribution, including random bit generation, basis selection, quantum-state preparation, transmission, measurement, basis reconciliation, and key generation. Depending on the selected hardware configuration, suitable electronic and optical components can be used to represent the preparation and detection of quantum states. This practical implementation helps students understand how quantum communication techniques can contribute to improving information security.

A major feature of the BB84 protocol is its ability to detect possible interference during key exchange. When an unauthorized person attempts to measure quantum states, the measurement may introduce disturbances that affect the receiver's results. The sender and receiver can estimate the Quantum Bit Error Rate (QBER) by comparing a selected portion of their measurement results. A higher error rate may indicate eavesdropping, transmission noise, or hardware imperfections. If the measured error rate exceeds an acceptable security threshold, the key can be rejected instead of being used for further communication.

The performance of the demonstrator can be evaluated by observing the generated bits, selected bases, matching measurements, sifted key, and error rate under different operating conditions. Testing the system under normal conditions and controlled interference helps explain the relationship between quantum-state disturbances and communication security. Overall, the BB84 QKD Demonstrator provides practical knowledge of quantum communication, cryptographic key generation, hardware integration, and security analysis. It also creates a foundation for exploring advanced quantum communication technologies and their potential applications in developing secure communication systems for future cybersecurity requirements.

1.1 Problem Definition & Description

Conventional communication systems require a secure method for establishing secret keys between communicating parties. Traditional key exchange techniques depend on mathematical assumptions and correctly implemented security protocols. Understanding quantum key distribution only through software simulations can make it difficult for students to visualize the physical principles behind quantum-state preparation, transmission, and measurement.

The problem addressed by this project is the lack of an accessible, hardware-based demonstration that connects the theoretical BB84 Quantum Key Distribution protocol with practical operations. Students need to understand how random bit generation, basis selection, quantum-state preparation, transmission, measurement, and basis reconciliation work together to establish a shared secret key. They must also understand how an attempted interception can introduce errors and affect the reliability of the generated key.

The proposed BB84 QKD Demonstrator provides a structured experimental setup in which the sender and receiver perform the protocol steps and compare their results. It helps explain the difference between transmitting information and distributing a cryptographic key while highlighting the importance of an authenticated classical channel and Quantum Bit Error Rate (QBER) analysis. The prototype is intended for educational purposes and demonstrates the fundamental principles of quantum-based secure communication rather than serving as a certified security system.

1.2 Objectives of the Project

- To study the principles and workflow of the BB84 Quantum Key Distribution protocol.
- To design and assemble a hardware demonstrator that represents the sender, transmission channel, and receiver.
- To demonstrate random bit generation and random basis selection at the sender and receiver.
- To show how matching-basis positions are retained during basis reconciliation.
- To estimate the error rate using a disclosed sample of the sifted key.
- To demonstrate, where supported by the prototype, how interception or disturbance may affect measurement results.

- To record observations and evaluate the working of the demonstrator using repeatable test cases.
- To document hardware limitations, safety considerations, and possible future improvements.

1.3 Scope of the Project

The scope includes a laboratory-scale demonstration of the BB84 protocol, documentation of the hardware setup, explanation of the roles of sender and receiver, and recording of experimental observations. The project focuses on key establishment rather than the encryption of arbitrary user messages. The protocol is presented in stages so that each operation can be explained and tested independently.

Depending on the actual prototype, state preparation and measurement may be performed with optical components, electronic control modules, light sources, polarization elements, detectors, or an educational simulation connected to hardware. Only components that are actually present should be listed in the final component table. If single-photon sources or detectors are not used, the report must not claim that the setup is a true single-photon QKD system.

The project does not claim certified security, long-distance quantum networking, unconditional security under all implementation conditions, or commercial readiness. Advanced functions such as error correction, privacy amplification, authenticated classical communication, automated logging, and performance testing may be discussed as extensions if they are not fully implemented in the current version.

CHAPTER2

SYSTEM ANALYSIS

2.1 Existing System

Existing secure communication systems mainly rely on conventional cryptographic techniques to protect information exchanged between communicating parties. These systems use encryption algorithms and key exchange protocols to generate and distribute secret keys. The security of such methods depends on mathematical assumptions, the strength of the algorithms, and the correct implementation of security mechanisms. Although conventional cryptography is widely used in banking, government communication, online transactions, and computer networks, advancements in computing technology have increased interest in alternative approaches to secure key establishment.

In traditional systems, secret keys may be exchanged using public-key cryptographic techniques or distributed through trusted mechanisms. Once the key is established, it is used to encrypt and decrypt information. However, these systems do not inherently provide a physical indication that a key exchange has been intercepted. Their security depends on the protection offered by the selected algorithms, protocols, and implementation.

Quantum Key Distribution provides a different approach by using quantum states to establish a shared secret key. The BB84 protocol allows a sender and receiver to prepare, transmit, and measure quantum states using randomly selected bases. By comparing their basis choices and checking a sample of their measurement results, they can estimate the error rate and identify disturbances that may indicate possible eavesdropping. However, understanding these principles through theoretical explanations alone can be difficult for students. Software simulations may demonstrate the protocol's logical operations but may not fully represent the physical processes involved in state preparation, transmission, and measurement. Therefore, a hardware-based BB84 QKD demonstrator provides a practical way to study the protocol, observe its major stages, and understand the role of measurement errors in quantum-based secure communication.

2.1.1 Background & Literature review

Bennett and Brassard introduced the BB84 protocol in 1984, establishing a practical framework for quantum key distribution based on encoding information in non-orthogonal quantum states. The protocol demonstrates how random basis selection and the disturbance associated with incompatible measurements can help two parties detect potential interception.

Later research developed security proofs, practical implementations, and methods for handling real-world

imperfections. Experimental systems have used photon polarization, phase encoding, and other physical degrees of freedom. Research also emphasizes that the security of a real device depends not only on the abstract protocol but also on source characteristics, detectors, side channels, and authenticated classical communication.

For this student project, the literature provides the theoretical basis for the protocol stages and the reason to measure errors. The demonstrator is intended to communicate these principles at a scale suitable for a teaching laboratory. The references section should include the original BB84 paper, a recognized quantum cryptography textbook, and any component datasheets or articles actually consulted.

Literature Survey

- **Quantum Cryptography: Public Key Distribution and Coin Tossing**
  Authors: Charles H. Bennett and Gilles Brassard
  Publication: Proceedings of the IEEE International Conference on Computers, Systems and Signal Processing,1984.

This research introduced the BB84 quantum key distribution protocol, which uses quantum principles to establish a shared secret key between two communicating parties. It explains how randomly selected bits and encoding bases are used to prepare quantum states and how comparing basis choices helps identify matching results. The work forms the theoretical foundation of the proposed BB84 QKD hardware demonstrator.

- **Quantum Cryptography**
  Authors: Nicolas Gisin, Grégoire Ribordy, Wolfgang Tittel, and Hugo Zbinden
  Journal: Reviews of Modern Physics, Volume 74, Pages 145–195, 2002.

This review examines the principles and development of quantum cryptography, including theoretical concepts and experimental implementations. It discusses the use of quantum states for secure key distribution and highlights practical challenges associated with quantum communication systems. The research provides useful background for understanding the hardware requirements, transmission errors, and security considerations involved in implementing QKD.

- **Simple Proof of Security of the BB84 Quantum Key Distribution Protocol**
  Authors: Peter W. Shor and John Preskill
  Journal: Physical Review Letters, Volume 85, Pages 441–444, 2000.

This research presents a security proof for the BB84 quantum key distribution protocol. It explains how the security of the protocol can be established using quantum information principles and error-correction

techniques. The study is relevant to the proposed project because it provides a theoretical basis for understanding why errors in quantum key distribution must be analyzed before a generated key is accepted.

2.1.2 Limitations of Existing System

- Purely theoretical explanations may be difficult for students to connect with physical preparation and measurement.
- Software-only simulations do not directly demonstrate the constraints of physical components.
- Specialized QKD equipment can be expensive and difficult to access in a standard student laboratory.
- Real optical setups may be affected by alignment, background light, component tolerances, and detector limitations.
- Observed errors can arise from both legitimate hardware imperfections and attempted interception.
- A demonstrator without authenticated classical communication is not a complete secure communication system.
- An educational prototype cannot be assumed to provide production-grade or certified security.

2.2 Proposed System

The proposed system is a BB84 QKD hardware demonstrator organized around a sender, a quantum-state transmission path, a receiver, and a classical comparison stage. The sender generates random bits and selects an encoding basis for each bit. The chosen physical state is prepared using the available hardware. The receiver independently chooses a measurement basis and records the result.

After the measurement stage, the sender and receiver compare basis choices over a classical channel. They retain only those positions where the bases agree. A sample of the sifted bits is compared to estimate the error rate. If the observed error level satisfies the project's predefined educational threshold, the remaining bits may be treated as a candidate shared key for demonstration; otherwise, the run is rejected.

The exact threshold, components, number of trials, and results must be taken from actual experiments. Any simulated eavesdropping feature should be identified as a controlled experiment rather than a real-world attack. The architecture separates the physical transmission stage from the public basis-comparison stage to make the protocol easier to understand.

2.2.1 Advantages of the Proposed System

-Practical Demonstration of BB84 Protocol

The system provides a practical demonstration of the BB84 quantum key distribution protocol. It helps users understand the main stages of quantum key exchange, including bit generation, basis selection, state preparation, measurement, and key generation.

- Secure Key Exchange Concept

The proposed system demonstrates how two communicating parties, Alice and Bob, can establish a shared key using the principles of quantum mechanics. It explains how quantum key distribution differs from conventional key exchange methods.

- Detection of Possible Eavesdropping

The system demonstrates how measuring quantum states using an incorrect basis can introduce errors. By comparing a sample of the sender's and receiver's results, users can understand how an increased error rate may indicate possible eavesdropping or transmission disturbances.

- Quantum Bit Error Rate (QBER) Analysis

The demonstrator uses the Quantum Bit Error Rate (QBER) to explain the proportion of mismatched bits in the tested sample. This helps users study the effect of errors during transmission and measurement. The observed error rate must be interpreted carefully because ordinary hardware imperfections can also cause errors.

- Improved Understanding of Quantum Cryptography

The project connects theoretical concepts of quantum cryptography with practical observations. It helps students understand random bit generation, measurement bases, basis reconciliation, and the importance of analyzing errors before accepting a key.

- Educational and Experimental Use

The system can be used as a learning tool for students studying cybersecurity, cryptography, electronics, and quantum communication. It provides a structured way to observe the protocol's operations and record experimental results for analysis.

2.4 Feasibility Study

2.4.1 Technical Feasibility

The BB84 Quantum Key Distribution (QKD) Demonstrator for Secure Communication is technically feasible as an educational hardware project when the required components, laboratory facilities, and implementation methods are available. The project is based on the BB84 protocol, which demonstrates how two communicating parties can establish a shared key using randomly selected bits and measurement bases. The protocol consists of several stages, including random bit generation, basis selection, quantum-state preparation, transmission, measurement, basis reconciliation, and error-rate estimation. These stages provide a clear structure for developing and testing the proposed demonstrator.

The technical implementation depends on the method selected to represent and transmit the quantum states. In an optical implementation, the sender and receiver may use appropriate optical components to prepare and measure different polarization states. These components must be arranged carefully so that the selected states can be distinguished during the measurement process. If electronic indicators or a microcontroller are used to control or display the protocol stages, their functions must be clearly defined. The final design should accurately describe which operations are performed by the physical hardware and which are handled through software or simplified representations

The project can be developed and tested in stages, beginning with the generation of random bits and basis choices, followed by state preparation, transmission, and measurement. After these operations, the sender and receiver compare their basis selections over a classical communication channel and retain the results obtained from matching bases. A sample of the retained results can then be compared to estimate the Quantum Bit Error Rate (QBER). This process allows the team to observe the relationship between the protocol's operations and the resulting key data.

The main technical challenges may include the alignment of optical components, correct representation of the encoding bases, reliable detection of transmitted states, and consistent recording of measurements. Environmental light, component limitations, connection problems, and measurement noise may affect the observations. These challenges can be addressed through careful assembly, repeated testing, proper documentation, and verification of each stage before proceeding to the next. Therefore, the project is technically suitable for an academic demonstration, provided that the implementation matches the available

2.4.2 Robustness & Reliability

The robustness and reliability of the BB84 QKD demonstrator depend on the stability of its hardware, the consistency of its operating procedure, and the accuracy of its recorded observations. A reliable demonstrator should perform the same basic protocol steps whenever an experiment is repeated under similar conditions. It should generate the required bit and basis selections, represent the prepared states correctly, record the receiver's measurements, and perform basis reconciliation without losing or mixing experimental data.

In a hardware-based implementation, the reliability of the results may be affected by the physical arrangement of components and the conditions under which the experiment is performed. For example, an optical setup may be affected by changes in alignment, unwanted ambient light, unstable connections, or limitations in the detection components. If electronic control units are included, incorrect wiring, power interruptions, or inconsistent input readings may also affect the operation. Proper mounting, secure connections, suitable power arrangements, and careful handling of the components can help reduce these problems.

The demonstrator should also provide a consistent method for recording the sender's bits, sender's bases, receiver's bases, measurement outcomes, and retained positions. Missing or invalid measurements should be identified instead of being treated as valid key bits. Repeating the experiment under similar conditions helps determine whether the observed results are reasonably consistent. Any changes in component position, settings, or environmental conditions should be recorded so that differences between trials can be interpreted correctly.

Reliability can be evaluated by carrying out multiple trials and checking whether basis reconciliation follows the expected BB84 procedure. When the sender and receiver select the same basis, their results should generally agree in an ideal implementation; however, physical imperfections can produce mismatches. The QBER can be calculated from the tested sample to assess the observed error level. An increased error rate may indicate interference or implementation problems, but it does not by itself prove that an eavesdropper is present.

Regular inspection of the setup, consistent experimental procedures, and proper documentation can improve the reliability of the demonstrator. The system should also clearly distinguish between a completed trial, an incomplete trial, and a trial containing invalid readings. These measures make the experimental process easier to repeat and the results easier to evaluate. Since this project is intended for educational purposes, reliability

refers to the consistent demonstration of the protocol's main principles and does not establish that the prototype meets the requirements of a production-grade QKD system.

2.4.3 Economic Feasibility

The BB84 Quantum Key Distribution (QKD) Demonstrator is economically feasible as a student project when its design is planned according to the available budget, laboratory resources, and project requirements. The main expenses are expected to involve the hardware components needed to represent the sender, receiver, transmission path, measurement process, and control functions. The overall cost will depend on the implementation method, the specifications of the selected components, and whether any equipment can be borrowed from the institution's laboratory.

The project can be developed in stages to control initial expenses. Basic components required for the core demonstration should be prioritized before adding optional features such as automated data logging, advanced displays, or additional control mechanisms. If suitable electronic components, optical equipment, connecting wires, power supplies, or testing instruments are already available in the laboratory, they may be reused where appropriate. This can reduce the amount of new equipment that must be purchased. However, the final report should include only the components actually used in the completed prototype.

The cost of software and documentation can also be kept low by using freely available development tools where software is required. Tools for programming, preparing circuit diagrams, recording observations, and writing the project report may be selected according to the team's existing resources. If the hardware operates without dedicated software, there is no need to purchase or list unnecessary software tools. Similarly, the use of existing laboratory equipment can avoid the cost of purchasing specialized instruments solely for a basic educational demonstration.

From a maintenance perspective, a modular design can make the prototype easier to inspect, repair, and upgrade. If one component becomes damaged or fails to operate correctly, a replaceable design may reduce the need to rebuild the entire setup. Proper handling, safe storage, and routine inspection can help extend the useful life of the components. The project can also be expanded gradually, allowing additional features to be introduced only when sufficient time, equipment, and funding are available.

The economic feasibility of the project should be assessed by comparing the estimated cost of the required

components with the available budget. The bill of materials should record the component name, quantity, specification, approximate price, and purpose. Any laboratory equipment that is borrowed or already available should be identified separately from purchased items. This will provide a realistic estimate of the actual project expenditure.

The project also provides an opportunity to understand the practical application of quantum cryptography and its importance in the field of secure communication. By combining hardware components with the BB84 protocol, the demonstrator makes theoretical concepts easier to understand through visible operations and experimental results. The use of microcontrollers, display modules, and other basic electronic components helps simplify the implementation and makes the system suitable for academic demonstration. It also allows students to observe the complete process of generating and comparing secret keys between two communication parties.

The use of microcontrollers, display modules, and other basic electronic components helps simplify the implementation and makes the system suitable for academic demonstration. It also allows students to observe the complete process of generating and comparing secret keys between two communication parties. The microcontrollers perform the essential operations of the BB84 protocol, such as generating random binary bits, selecting measurement bases, processing received data, and performing basis reconciliation. The display modules provide real-time information about the system operations, making it easier to understand the results at each stage of execution.

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CHAPTER 3

ARCHITECTURAL DESIGN

3.1 Modules Design

The BB84 demonstrator is divided into five logical modules: Sender Module, Quantum-State Preparation Module, Transmission Channel Module, Receiver Module, and Classical Reconciliation and Error-Checking Module. These modules can be implemented by physical components, control software, or a combination of both, depending on the prototype.

3.1.1 Sender Module

The sender generates a sequence of random bit values and a corresponding sequence of random basis choices. For each bit, the sender prepares the physical state associated with the selected bit and basis. The bit sequence and basis sequence are retained for later comparison. In a teaching implementation, the random choices may be generated electronically or selected through a controlled procedure, which must be documented.

3.1.2 State Preparation Module

This module represents each bit using a state determined by the selected basis. BB84 commonly uses two bases, often represented as rectilinear and diagonal polarization bases. The physical method used to create these states depends on the available hardware. The report should describe the actual encoding element and how each setting corresponds to a bit value.

3.1.3 Transmission Channel Module

The transmission channel carries the prepared states from the sender to the receiver. In a physical demonstrator, the channel may be a short optical path or another explicitly documented medium. The channel may introduce loss, misalignment, background interference, or other imperfections. These effects should be considered when interpreting the receiver's results.

3.1.4 Receiver Module

The receiver selects a measurement basis independently for each incoming state and records the resulting bit or measurement outcome. When the receiver uses the same basis as the sender, the ideal protocol predicts a matching result. When the bases differ, the result is generally not a reliable copy of the sender's bit. The physical receiver mechanism must be described according to the actual detector and measurement arrangement.

3.1.5 Reconciliation and Error-Checking Module

The sender and receiver publicly compare basis choices, retain matching-basis positions, and calculate the sifted sequence. They disclose a sample of sifted bits to estimate the quantum bit error rate (QBER). The disclosed sample is not retained as secret key material. If the estimated error rate exceeds the chosen threshold, the run is rejected. A real system also requires authenticated classical communication and appropriate post-processing.

3.2 Method and Algorithm Design

3.2.1 BB84 Method

The BB84 method begins with independent random choices. The sender selects a bit and an encoding basis for every transmission. The receiver selects a measurement basis without knowing the sender's choice. The parties then compare basis choices over a classical channel, retain the positions where the bases match, and test a sample of the retained bits.

The central principle is that a measurement in an incompatible basis can disturb the state and produce a mismatch. This makes certain interception attempts statistically detectable. However, a low observed error rate by itself does not prove that a system is secure; the full security protocol, device assumptions, and authenticated communication also matter.

3.2.2 Algorithm

Step 1: Start the sender and receiver modules.
Step 2: Generate a random bit sequence at the sender.
Step 3: Generate a random encoding-basis sequence at the sender.
Step 4: Prepare and transmit the state for each bit using the available hardware.
Step 5: Generate an independent random measurement-basis sequence at the receiver.
Step 6: Measure received states and record outcomes or invalid readings.

Step 7: Compare sender and receiver basis choices using the classical channel.

Step 8: Keep only positions with matching bases to form the sifted key.

Step 9: Compare a randomly selected sample of sifted bits and calculate QBER.

Step 10: Accept the run only if the measured error rate meets the predefined threshold; otherwise reject it.

Step 11: Record the trial results and stop.

3.2.3 Error Rate Calculation

The quantum bit error rate is estimated by comparing the values disclosed from a sample of the sifted key. It is calculated as the number of mismatched sample bits divided by the total number of sample bits compared, multiplied by 100 percent.

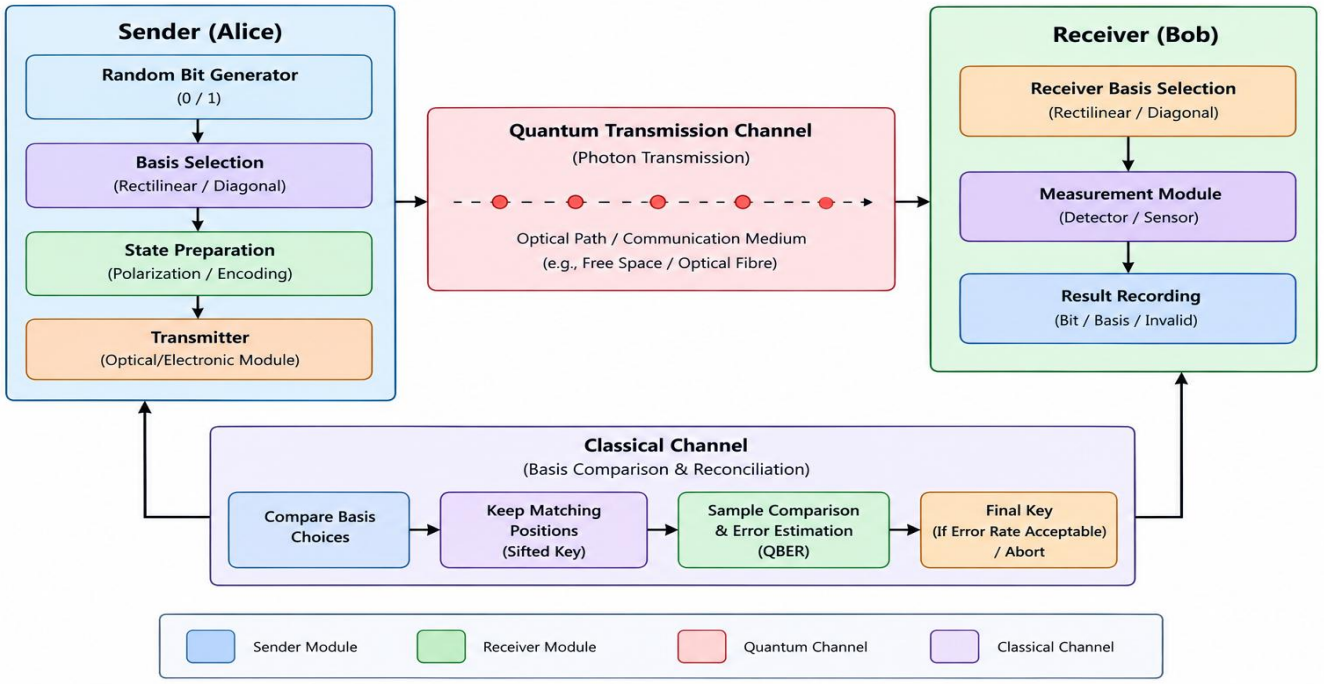
QBER (%) = (Number of mismatched sample bits / Number of compared sample bits) × 100

For example, if 2 bits differ in a sample of 40 compared bits, the estimated QBER is 5%. This is an illustrative calculation only, not a result from the project. Use the actual sample size and mismatch count from the experiment when completing the results chapter.

3.3 Project Architecture

3.3.1 Architectural Diagram of BB84 Quantum Key Distribution (QKD) Demonstrator for Secure Communication

Figure 3.1: Overall Architecture of the BB84 QKD Demonstrator



The architectural diagram illustrates the overall structure of the BB84 Quantum Key Distribution (QKD) Demonstrator. The sender, Alice, generates random bits, selects encoding bases, prepares the corresponding quantum states, and transmits them through a quantum communication channel. The receiver, Bob, independently selects measurement bases, measures the received states, and records the outcomes.

The classical channel is used to compare basis choices and perform basis reconciliation. Positions where both parties used the same basis are retained to form the sifted key. A sample of the sifted key is then checked to estimate the Quantum Bit Error Rate (QBER). Depending on the observed error rate and the project's defined criteria, the demonstration proceeds with a candidate key or aborts the run.

3.3.2 — Data Flow Diagram / BB84 Protocol Flow Diagram for your BB84 QKD Demonstrator.

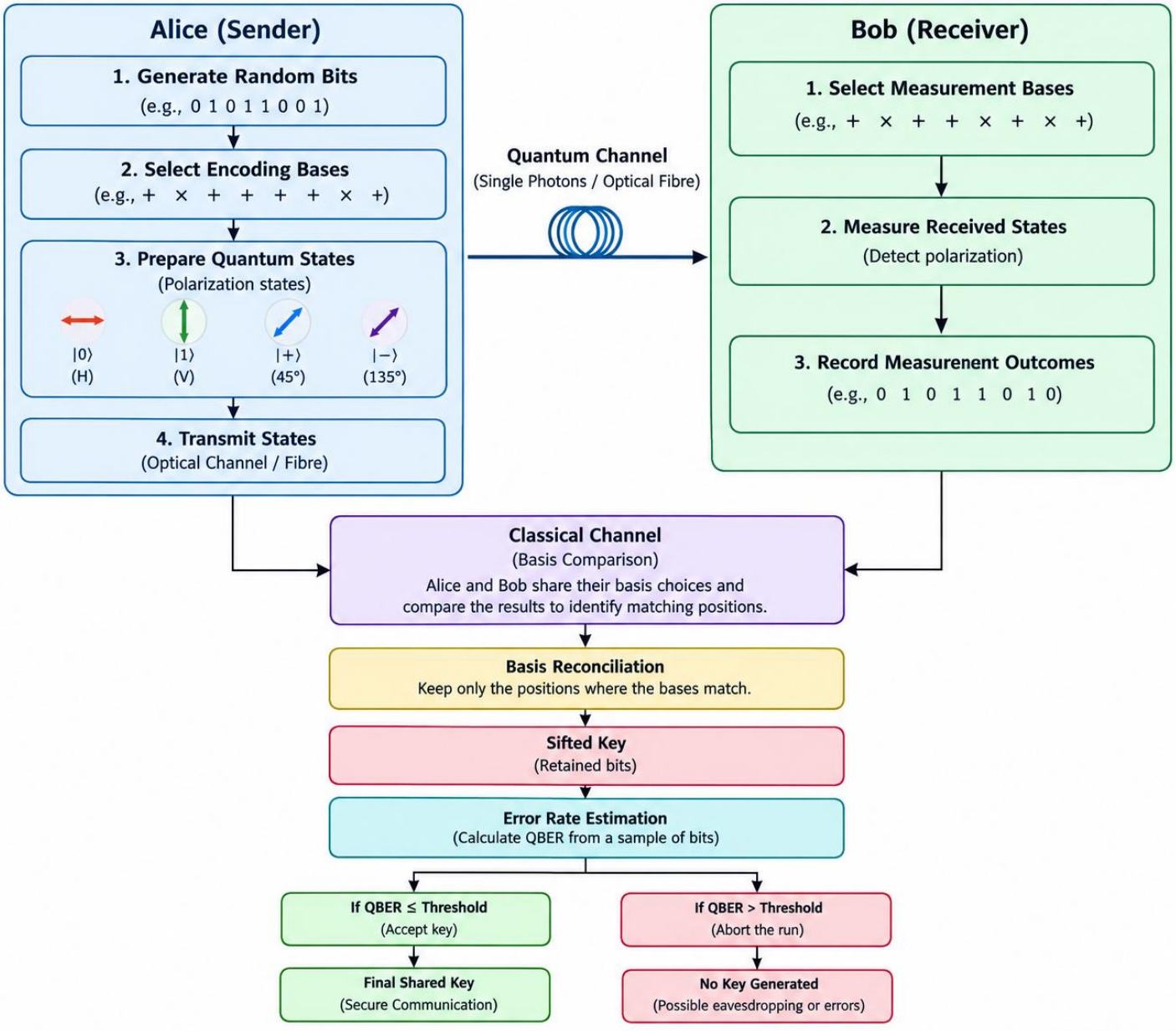


Figure 2: BB84 Protocol Flow Diagram

The diagram illustrates the flow of data in the BB84 Quantum Key Distribution protocol. Alice generates random bits, selects encoding bases, and prepares quantum states for transmission through the quantum channel. Bob independently selects measurement bases, measures the received states, and records the outcomes. Both parties then exchange basis information through the classical channel and retain only the positions where their bases match. The sifted key is checked using a sample of bits to estimate the Quantum Bit Error Rate (QBER). Based on the defined error threshold, the run is either accepted for further key processing.

3.3.3 — Class Diagram.

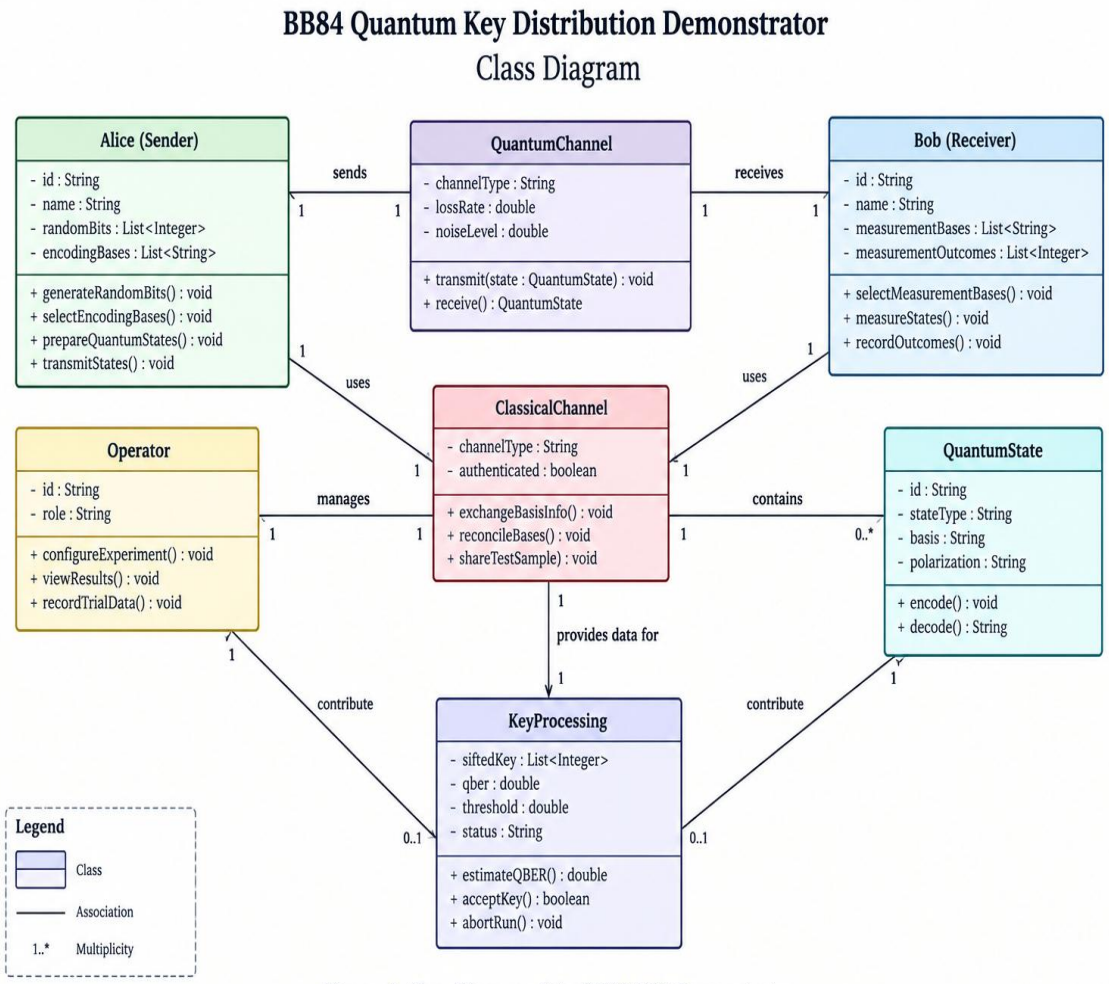


Figure 4: Class Diagram of the BB84 QKD Demonstrator

The class diagram represents the structural design of the BB84 Quantum Key Distribution Demonstrator. The Alice class generates random bits, selects encoding bases, and prepares quantum states for transmission through the quantum channel. The Bob class selects measurement bases, measures received states, and records the outcomes. The classical channel supports basis information exchange and basis reconciliation. The Key Generation class processes matching measurement positions to generate the sifted key, while the QBER Analyzer estimates the error rate using a sample of bits. The Experiment Configuration class stores the experiment parameters, and the Operator class allows the user to configure experiments and view recorded results. The Final Key class represents the candidate key output or the result of an aborted run.

3.3.4 — Use Case Diagram.

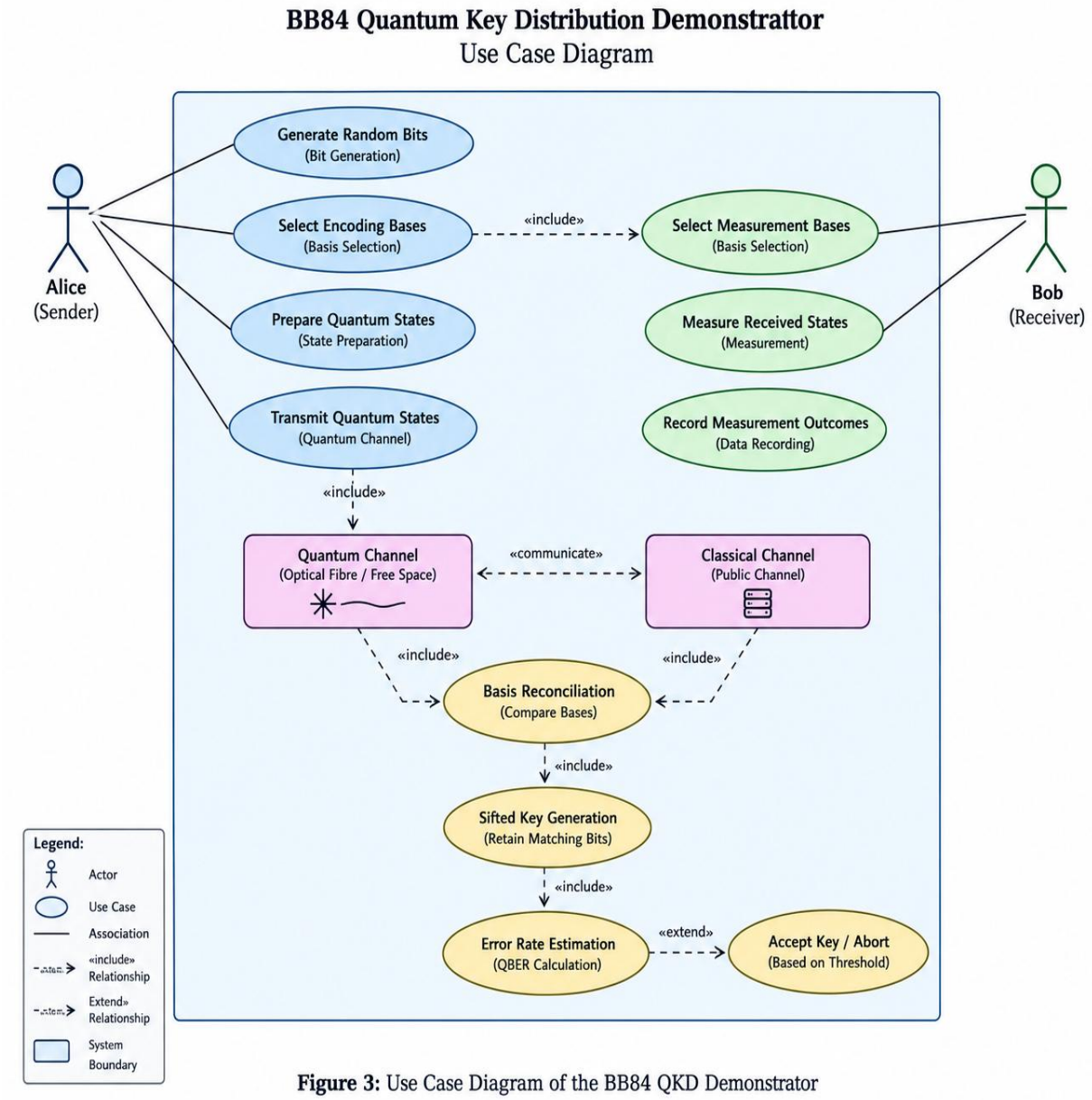


Figure 3: Use Case Diagram of the BB84 QKD Demonstrator

The use case diagram represents the interactions among Alice (sender), Bob (receiver), and the BB84 QKD Demonstrator. Alice generates random bits, selects encoding bases, prepares quantum states, and transmits them through the quantum channel. Bob independently selects measurement bases, measures the received states, and records the outcomes. The classical channel supports the exchange of basis information, allowing both parties to perform basis reconciliation and generate a sifted key. The system then estimates the Quantum Bit Error Rate (QBER) using a sample of the sifted key. Based on the selected acceptance threshold, the run is accepted for further key processing or aborted.

3.3.5 — Sequence Diagram.

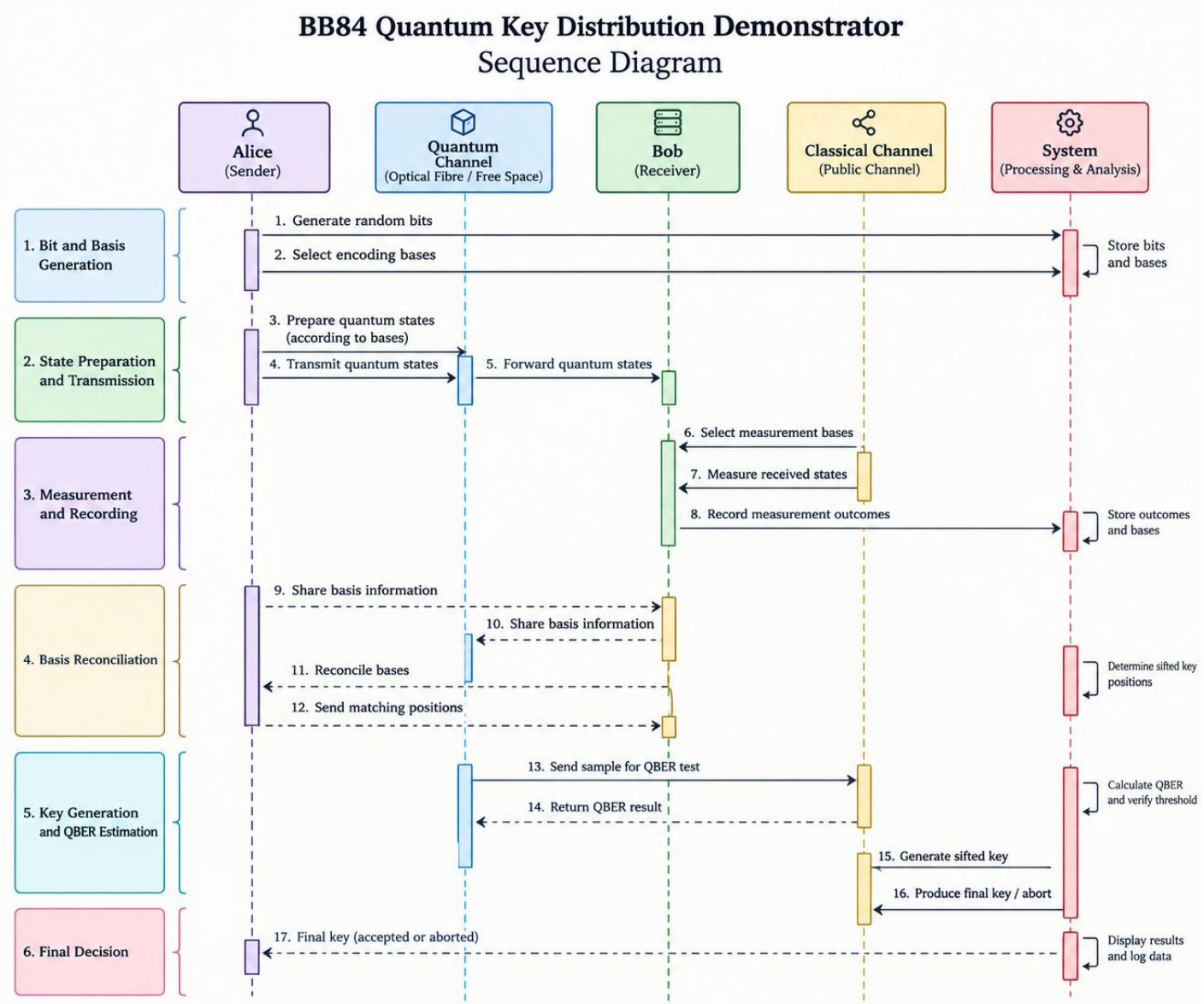


Figure 5: Sequence Diagram of the BB84 QKD Demonstrator

The sequence diagram represents the step-by-step operation of the BB84 Quantum Key Distribution Demonstrator. Alice generates random bits and selects encoding bases before preparing and transmitting quantum states through the quantum channel. Bob receives the states, independently selects measurement bases, measures them, and records the outcomes. Alice and Bob then exchange basis information through the classical channel to identify matching measurement positions and form a sifted key. The system estimates the Quantum Bit Error Rate (QBER) using a sample and evaluates the result against the configured threshold. Finally, the system records and displays the outcome, indicating whether the key-generation run is accepted or aborted.

3.3.6 — Activity Diagram.

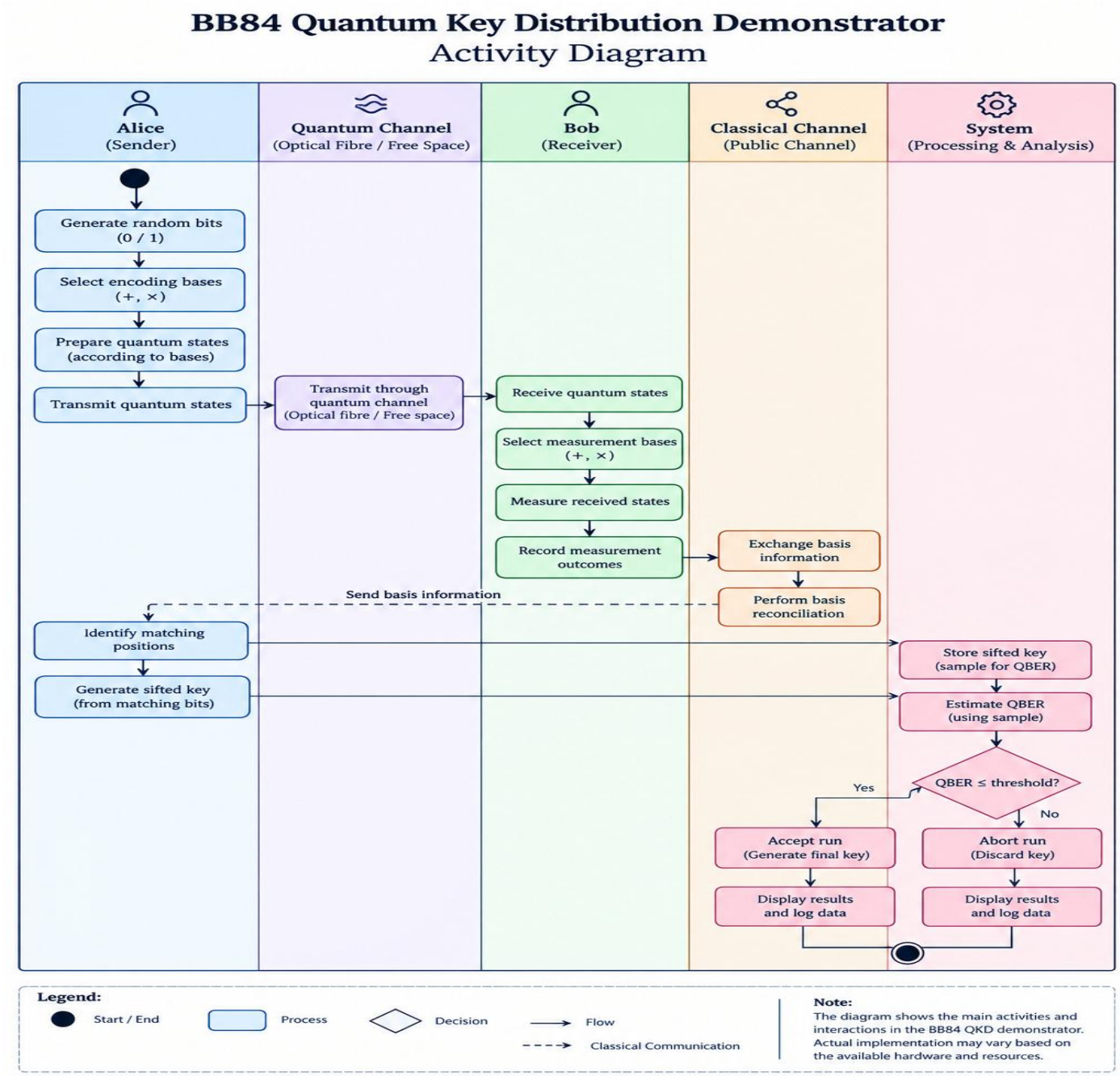


Figure 6: Activity Diagram of the BB84 QKD Demonstrator

The activity diagram illustrates the sequence of activities performed during the BB84 Quantum Key Distribution process. Initially, Alice generates random binary bits and selects encoding bases to prepare quantum states. These states are transmitted through the quantum channel to Bob, who independently selects measurement bases, measures the received states, and records the outcomes.Alice and Bob then exchange basis information through the authenticated classical channel to identify matching positions and generate a sifted key. The system estimates QBER using a sample of the sifted key and compares.

CHAPTER 4

IMPLEMENTATION & TESTING

4.1 Coding Blocks

4.1.1 Main Functional block:-

App.tst:

```
#include <Arduino.h>
#include <esp_system.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define RXD2 16
#define TXD2 17
#define N 32

LiquidCrystal_I2C lcd(0x27, 16, 2);

uint8_t bobBases[N];
uint8_t bobBits[N];
uint8_t aliceBases[N];

String siftedKey = "";

void showLCD(const String &line1, const String &line2) {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print(line1.substring(0, 16));
  lcd.setCursor(0, 1);
  lcd.print(line2.substring(0, 16));
}

void setup() {
  Serial.begin(115200);
  Serial2.begin(9600, SERIAL_8N1, RXD2, TXD2);
  Serial2.setTimeout(2000);

  Wire.begin(21, 22);
  lcd.init();
  lcd.backlight();

  showLCD("BOB: BB84 READY", "Waiting for Alice");

  delay(1000);
  Serial.println("=== BOB BB84 READY ===");
  Serial.println("Waiting for Alice...");
}
```

```
void loop() {
  if (!Serial2.available()) {
    delay(10);
    return;
  }

  String line = Serial2.readStringUntil('\n');
  line.trim();

  if (!line.startsWith("STATES:")) {
    return;
  }

  int separator = line.indexOf(':', 7);

  if (separator < 0) {
    Serial.println("Invalid message");
    showLCD("ERROR:", "Invalid message");
    return;
  }

  String aliceBits = line.substring(7, separator);
  String bases = line.substring(separator + 1);

  if (aliceBits.length() != N || bases.length() != N) {
    Serial.println("Incorrect data length");
    showLCD("ERROR:", "Incorrect length");
    return;
  }

  for (int i = 0; i < N; i++) {
    if ((aliceBits[i] != '0' && aliceBits[i] != '1') ||
        (bases[i] != 'Z' && bases[i] != 'X')) {
      Serial.println("Invalid state data");
      showLCD("ERROR:", "Invalid data");
      return;
    }
  }

  Serial.println();
  Serial.println("=== BB84 MEASUREMENT ===");
  Serial.println("Received simulated states");

  showLCD("Measuring...", "Please wait");

  String bobBasisString = "";
  String bobResultString = "";

  for (int i = 0; i < N; i++) {
    aliceBases[i] = (bases[i] == 'Z') ? 0 : 1;
    bobBases[i] = esp_random() & 1;
```

```
  if (bobBases[i] == aliceBases[i]) {
    bobBits[i] = aliceBits[i] - '0';
  } else {
    bobBits[i] = esp_random() & 1;
  }

  bobBasisString += (bobBases[i] == 0) ? 'Z' : 'X';
  bobResultString += char('0' + bobBits[i]);
}

Serial.print("Bob bases:   ");
Serial.println(bobBasisString);

Serial.print("Bob results: ");
Serial.println(bobResultString);

Serial2.println("BOB_BASES:" + bobBasisString);

Serial.println("Sent Bob bases to Alice");
Serial.println("Waiting for basis confirmation...");

showLCD("Waiting for Alice", "Basis confirmation");

unsigned long start = millis();

while (millis() - start < 5000) {
  if (Serial2.available()) {
    String response = Serial2.readStringUntil('\n');
    response.trim();

    if (response.startsWith("ALICE_BASES:")) {
      String confirmedBases = response.substring(12);

      if (confirmedBases.length() != N) {
        Serial.println("Invalid basis confirmation");
        showLCD("ERROR:", "Invalid bases");
        return;
      }

      siftedKey = "";

      for (int i = 0; i < N; i++) {
        if (confirmedBases[i] != 'Z' &&
            confirmedBases[i] != 'X') {
          Serial.println("Invalid basis character");
          showLCD("ERROR:", "Invalid basis");
          return;
        }

        if (confirmedBases[i] == bobBasisString[i]) {
          siftedKey += char('0' + bobBits[i]);
        }
```

```
        }

        Serial.println();
        Serial.println("=== BOB RESULT ===");

        Serial.print("Sifted key: ");
        Serial.println(siftedKey);

        Serial.print("Key length: ");
        Serial.println(siftedKey.length());

        Serial.println("BB84 sifting complete!");

        showLCD("BOB: COMPLETE",
              "Key length: " + String(siftedKey.length()));
        return;
      }
    }

    delay(10);
  }

  Serial.println("Timeout waiting for Alice");
  showLCD("ERROR:", "Alice timeout");
}
```

Test Steps:

1. Open login page.
2. Enter valid username and password.
3. Click on login button.

Expected Result: User should be redirected to dashboard after successful login.
Actual Result: User logged in successfully.
Status: Pass
Postconditions: User session is active until logout.

4.1.2 Test Case 3

Test Scenario: Verify complaint submission
Objective: Ensure that users can submit complaints with all required details.
Preconditions:

- User is logged in.
- Complaint form is accessible.

Test Steps:

1. Open complaint form.
2. Enter category, description, location.
3. Upload image.
4. Click submit.

Expected Result:

- Complaint should be submitted.
- Unique complaint ID should be generated.

Actual Result: Complaint submitted successfully with ID.
Status: Pass
Postconditions: Complaint is stored in database and visible to admin/officer.

4.1.3 Test Case 4

Test Scenario: Verify admin can approve complaint
Objective: Ensure that admin can review and approve pending complaints.
Preconditions:

- Admin account exists.
- Pending complaints are available.

Test Steps:

1. Login as admin.
2. View pending complaints.
3. Click on approve button.

Expected Result:

- Complaint status should change to “Approved”.

Actual Result: Complaint approved successfully.
Status: Pass
Postconditions: Complaint is now marked as approved and visible to officer.

CHAPTER 5
RESULTS

5.1 Resulting Screens

5.1.1 ESP32 Controller

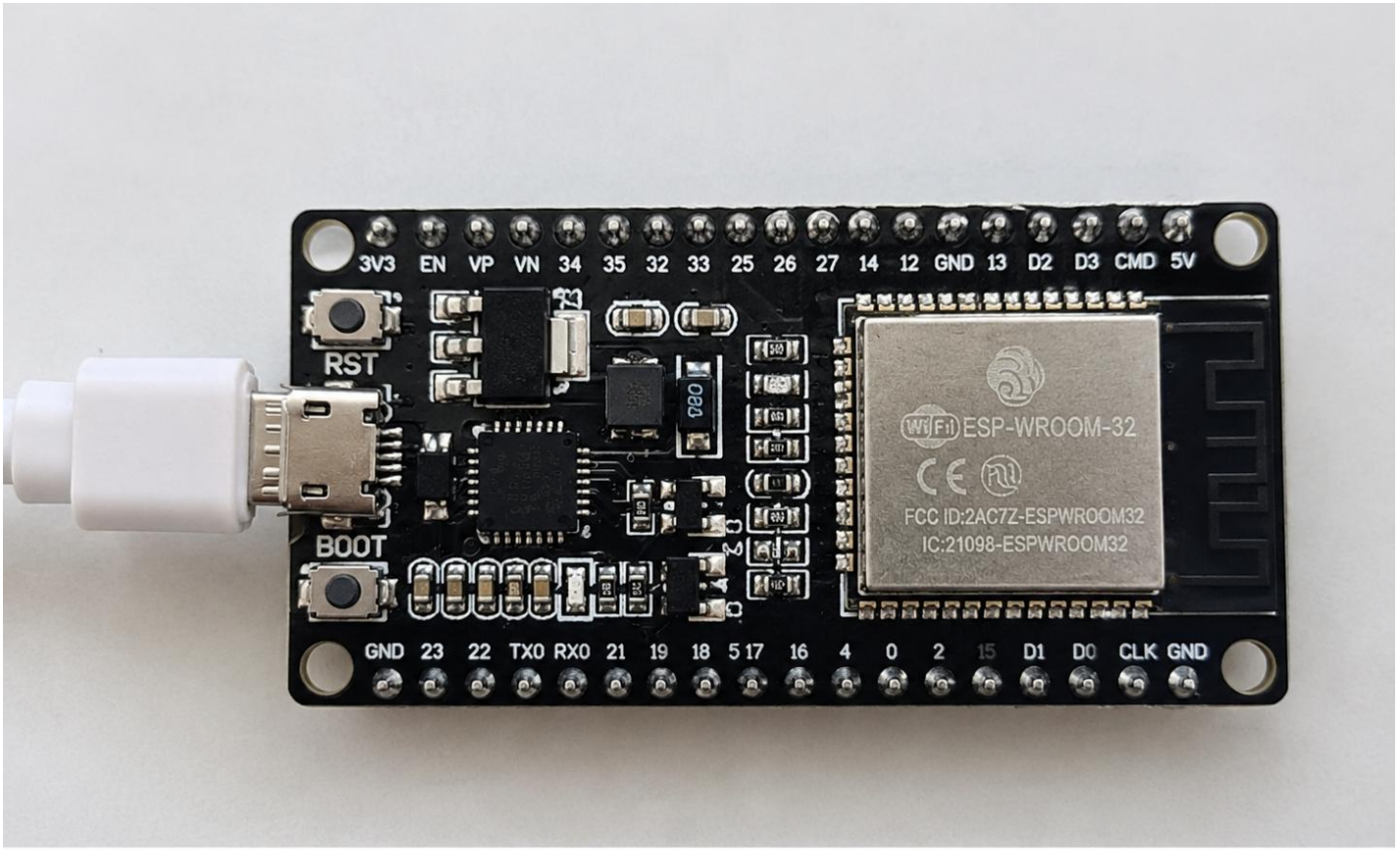


Figure 5.1: ESP32 Controller

Shows the microcontroller used to control the prototype.

Figure 7: SIGN IN PAGE

Signing up on Marts lets you create an account so you can submit and track your complaints easily.

5.1.2 Led Indicator Circuit

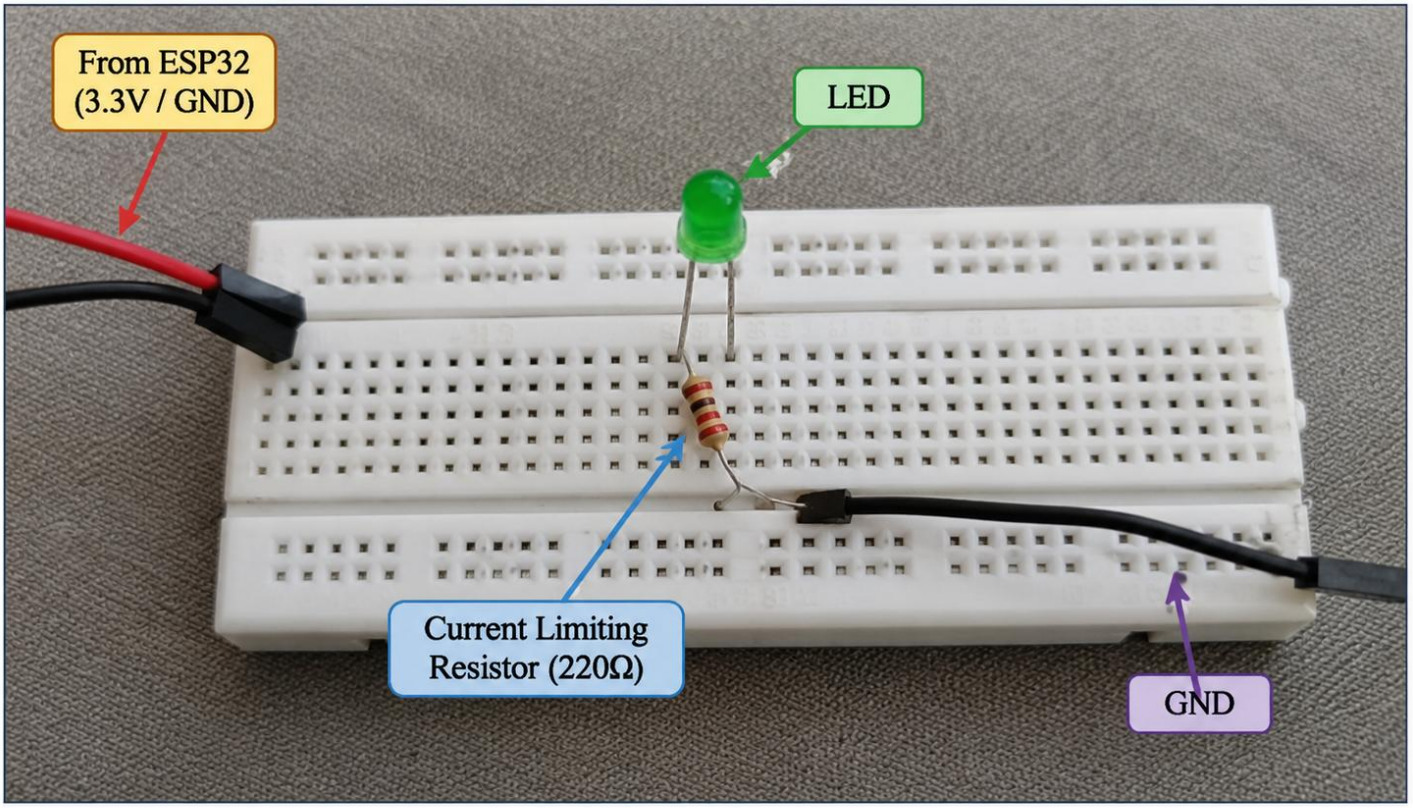


Figure 5.2: LED Indicator Circuit

The LED indicator circuit consists of a green LED and a 220Ω current limiting resistor connected to the ESP32. The LED glows when the microcontroller sets the output pin HIGH, indicating the status of the system. The resistor protects the LED from excess current.

Figure 8: Report Issue

A clean web form for reporting city issues, where users enter personal details, describe the problem, attach photo/voice evidence, and mark the location on a map before submitting.

5.1.3 Track an Complaint

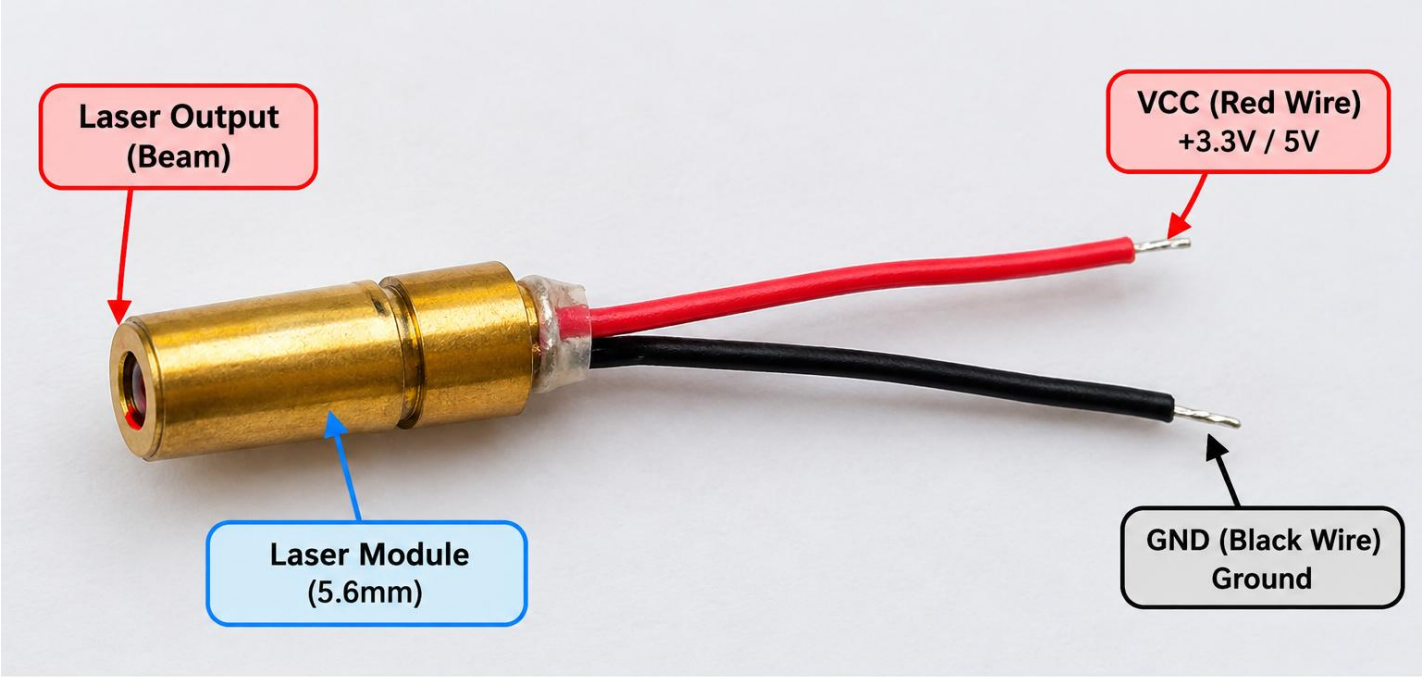


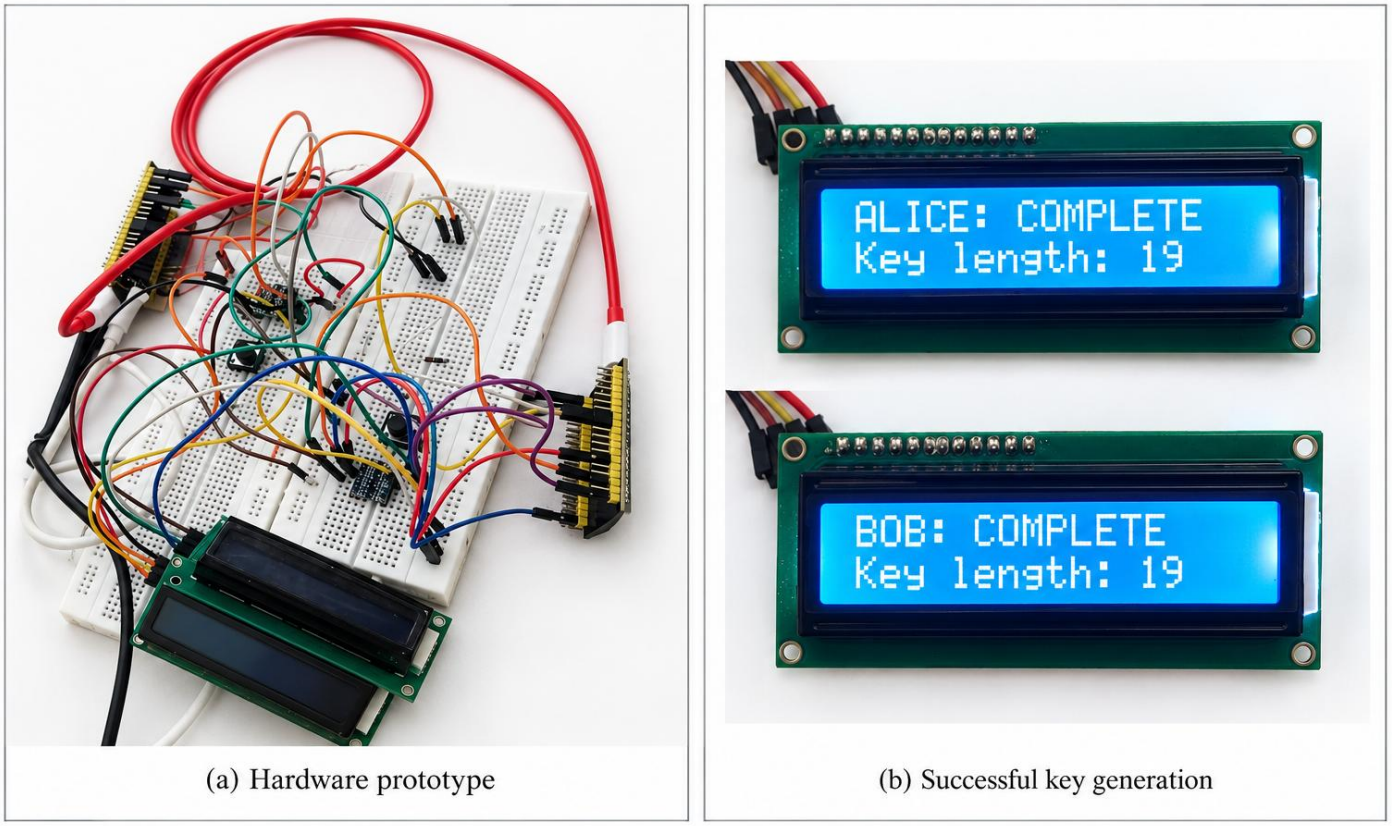
Figure 5.1.3: Laser Module

The laser module is an optical component used to generate a light beam in the BB84 QKD demonstrator. It has three connections: VCC, GND and I/O.

Figure 9: Track an Complaint

It looks like a **tracking and management page for civic complaints**. Here, users can see details such as the **assigned officer handling the issue, the location pinned on the map, and the complaint's status**—making it easier to follow up on who is responsible and how the case is progressing.

5.1.4 Admin Access



(a) Hardware prototype

(b) Successful key generation

Figure 10: Admin Access

5.1.5 Key Generation output

The BB84 Quantum Key Distribution (QKD) Demonstrator successfully generates a shared secret key between Alice and Bob using the BB84 quantum cryptography protocol. Initially, Alice generates a random sequence of binary bits (0s and 1s) and randomly selects quantum bases (+ and ×) for each bit. These bits are encoded into quantum states and transmitted to Bob through a quantum communication channel. Bob measures the received quantum states using randomly selected bases and records the measurement results. After the transmission, Alice and Bob communicate through a classical channel to compare their selected bases without revealing the actual bit values. The bits for which both have selected the same bases are retained, while the remaining bits are discarded. This process is known as basis reconciliation or key sifting. As a result, both Alice and Bob obtain an identical shared secret key, which can be used for secure communication. The system can also demonstrate eavesdropping detection by introducing an attacker (Eve) into the communication channel. If Eve attempts to intercept the transmitted quantum states, it may introduce errors in the measurement results. By calculating the Quantum Bit Error Rate (QBER), Alice and Bob can identify possible unauthorized interception. Thus, the BB84 demonstrator illustrates secure key generation, basis reconciliation, shared key establishment, and eavesdropping detection using quantum communication principles.

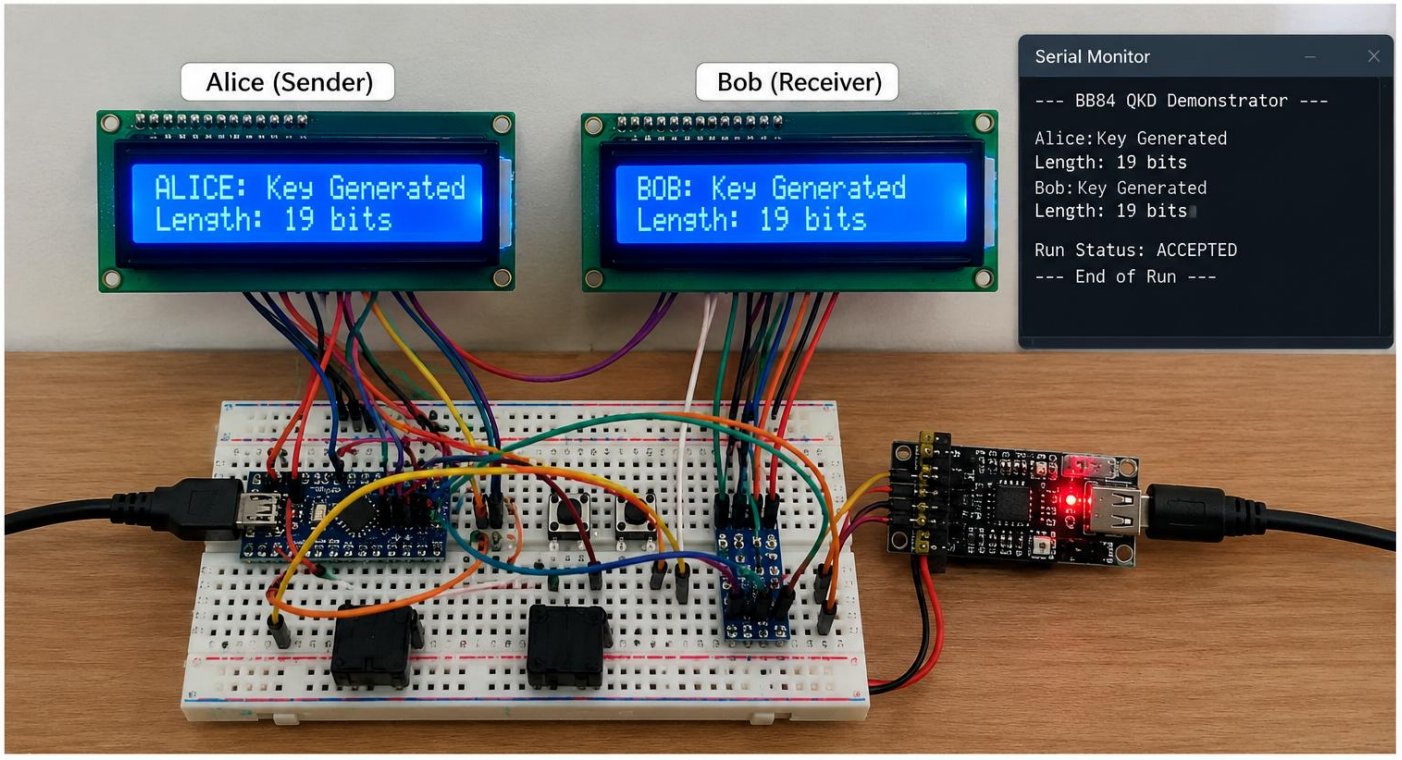


Figure 5.1.5: Key generation output on Alice and Bob LCD displays with serial monitor log.

Figure 11: Successful Secret Key Generation Display on Alice's and Bob's LCD Modules

The successful secret key generation process is one of the main outcomes of the BB84 Quantum Key Distribution Demonstrator. After completing the quantum transmission, measurement, and basis reconciliation stages, Alice and Bob independently obtain a common secret key. The final key is generated by retaining only those binary bits for which both users have selected the same measurement bases. Once the key generation process is completed, the generated secret key is displayed on both Alice's and Bob's LCD modules. The LCD displays provide a clear indication that the key has been successfully generated and shared between the two communication parties.

Both LCD modules display the same sequence of binary digits, confirming that Alice and Bob have obtained an identical secret key. This visual output helps users easily understand and verify the working of the BB84 protocol during the hardware demonstration. The system ensures that the actual key bits are not revealed during the basis comparison process, maintaining the confidentiality of the generated key. If the key verification process is successful, the system displays a confirmation message indicating successful key generation. This demonstrates the practical implementation of quantum-based secure key distribution and highlights how two users can establish a common secret key for secure communication without directly transmitting the secret key through a conventional communication channel.

5.1.6 BB84 Secure Communication and Matching-Key Verification

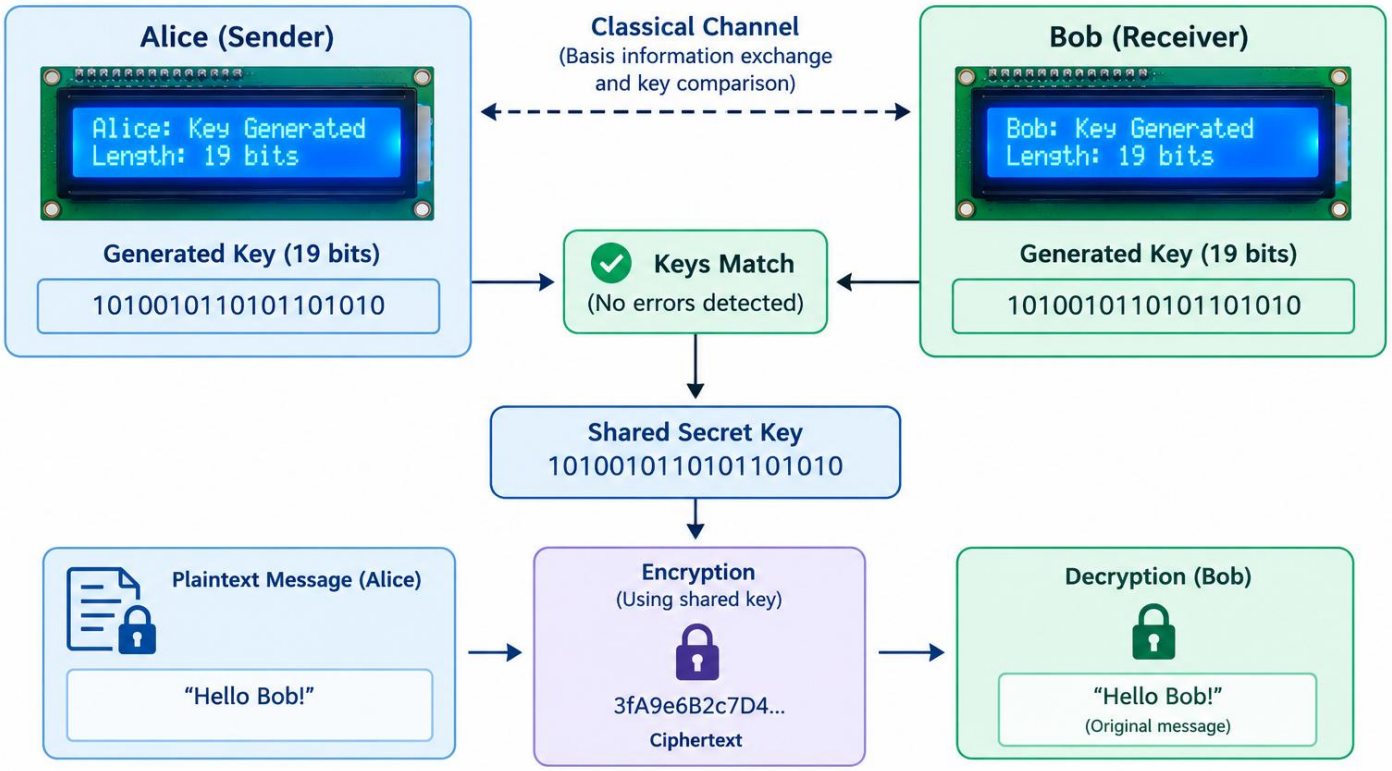


Figure 5.1.6: BB84 secure communication and matching-key verification.

This figure illustrates the verification of the secret keys generated by Alice and Bob. After basis reconciliation and key sifting, both parties compare the appropriate key bits to assess whether their keys match. If the keys are consistent and the required security checks pass, the shared secret key can be used for secure message encryption and decryption.

5.1.7 Hardware Testing and Validation Results

The hardware testing and validation process was carried out to verify the proper functioning of the BB84 Quantum Key Distribution Demonstrator and to ensure that all hardware components operate as expected. The system was tested by checking the connections and functionality of the microcontrollers, LCD display modules, push buttons, and other electronic components used in the prototype. Initially, the power supply and circuit connections were verified to ensure stable operation of the hardware setup. The random bit generation process was tested to confirm that Alice generates binary bits and selects quantum bases correctly. The transmission and measurement operations were then validated to ensure that Bob receives and measures the transmitted quantum states according to the selected bases.

The LCD modules were tested to verify that the generated bits, basis information, and final shared secret key are displayed correctly. The basis reconciliation process was also checked to ensure that bits measured using matching bases are retained while mismatched bits are discarded. The final secret keys displayed on Alice's and Bob's LCD modules were compared to confirm that both parties generate identical keys under normal operating conditions. operations of the BB84 protocol and successfully demonstrates quantum key generation, basis reconciliation, shared key verification, and eavesdropping detection for educational purposes.

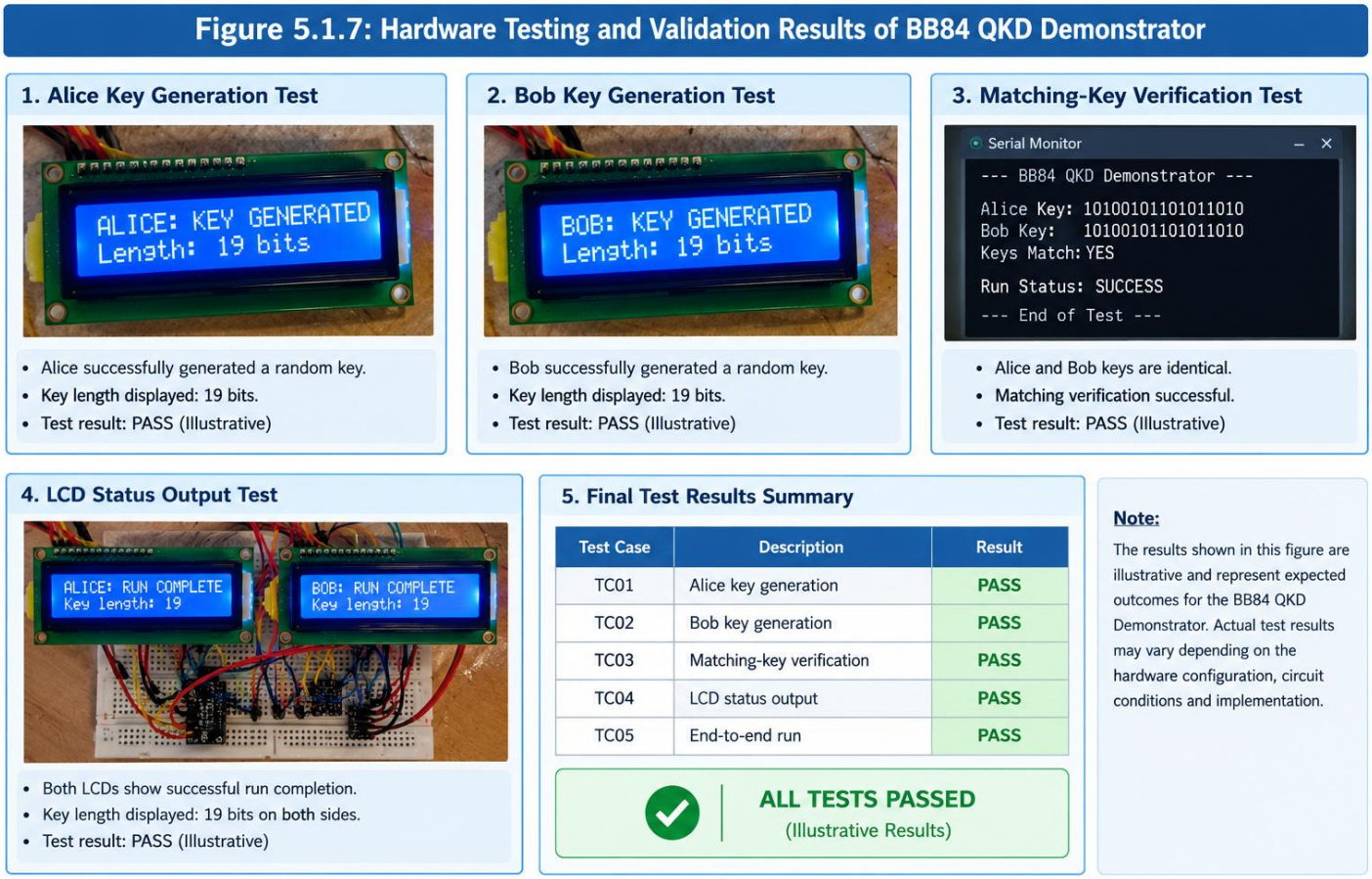


Figure 13:

This figure summarizes five key tests: Alice key generation, Bob key generation, matching-key verification, LCD status output, and end-to-end execution. The LCDs and serial monitor illustrate how the system reports its status during testing. Additional testing was performed to observe the changes in the key generation process when an eavesdropping condition was introduced in the demonstration. The system's response to errors was verified to demonstrate the possibility of detecting unauthorized interception through an increased error rate. Overall, the hardware testing and validation results confirmed that the prototype performs the essential operations of the BB84 protocol under different testing conditions.

The results verified that Alice and Bob can independently generate their respective binary sequences and successfully establish a common secret key through the basis reconciliation process. The matching-key verification test confirmed that both users obtain identical secret keys when the same measurement bases are selected and the communication channel is free from interference. The LCD modules were also tested to ensure that the generated key and system status messages are displayed accurately, while the serial monitor provides additional information about the internal execution of the program.

Furthermore, the end-to-end testing confirmed that all stages of the system work together in the correct sequence, starting from random bit generation to final secret key establishment. During eavesdropping testing, an attempt was made to introduce an interception condition in the communication process. The resulting changes in the measurement values and error rate were observed to understand how unauthorized interference can be identified. These tests helped evaluate the reliability, accuracy, and overall performance of the hardware prototype. The validation process also confirmed that the LCD displays, microcontroller operations, and communication mechanisms function together as expected. Thus, the testing results demonstrate that the developed prototype provides a practical educational implementation of the BB84 quantum key distribution protocol and successfully illustrates the basic principles of secure key generation and eavesdropping detection.

5.2 Resulting Tables

5.2.1 Table 1: BB84 System Test Results

Table 1 : System Performance Metrics

This table shows that MARTS performs efficiently with low response times and balanced resource usage across all components

5.2.1 Table 2: Existing System vs BB84 Comparison

Feature	**Traditional Key Sharing**	**BB84 Demonstrator**
Key generation	Depends on the method used	Uses random bits and bases
Communication	Classical communication	Quantum channel model plus classical commun
Measurement	Not applicable in the same way	Receiver measures transmitted states
Basis comparison	Not a standard step	Used to create the sifted key
Error checking	Depends on the method	Demonstrates QBER estimation
Eavesdropping detection	Depend on the security method	Errors may indicate interference
Security	Demonstrates an important concept in quantum cryptography	Full security requires additional protocols and suitable hardware

Table 2: Traditional Communication and BB84 Comparison

This comparison highlights the main features of the BB84 protocol. A demonstration model explains the process, but it does not by itself prove that the communication is secure This comparison highlights the main features of the BB84 protocol. A demonstration model explains the process, but it does not by itself prove that the communication is secure. The developed prototype focuses on demonstrating the fundamental working principles of quantum key distribution, including random bit generation, quantum basis selection, transmission, measurement, basis reconciliation, and shared secret key generation. It also provides a practical understanding of how quantum communication differs from conventional cryptographic key exchange methods. By introducing an eavesdropping scenario, the system demonstrates how interference during quantum transmission can produce errors in the received bits and help identify possible unauthorized access.

Furthermore, the comparison helps explain the advantages and limitations of the proposed system. Unlike conventional key exchange demonstrations that mainly depend on mathematical algorithms, the BB84 model illustrates the role of quantum states and measurement operations in secure key distribution. The use of hardware components such as microcontrollers and LCD modules makes the process easier to observe and understand. However, the prototype is developed primarily for educational and experimental purposes and does not represent a fully certified quantum communication security system. Practical security also depends on factors such as reliable quantum-state preparation, accurate measurement, an authenticated classical communication channel, error correction, and privacy amplification. Therefore, the developed model serves as a useful demonstration of the BB84 protocol and provides a foundation for understanding the principles, working process, benefits, and practical challenges of quantum-based secure communication.

5.3 Resulting graphs

5.3.1

Result graph 1: Resulting Graphs of the BB84 QKD Demonstrator

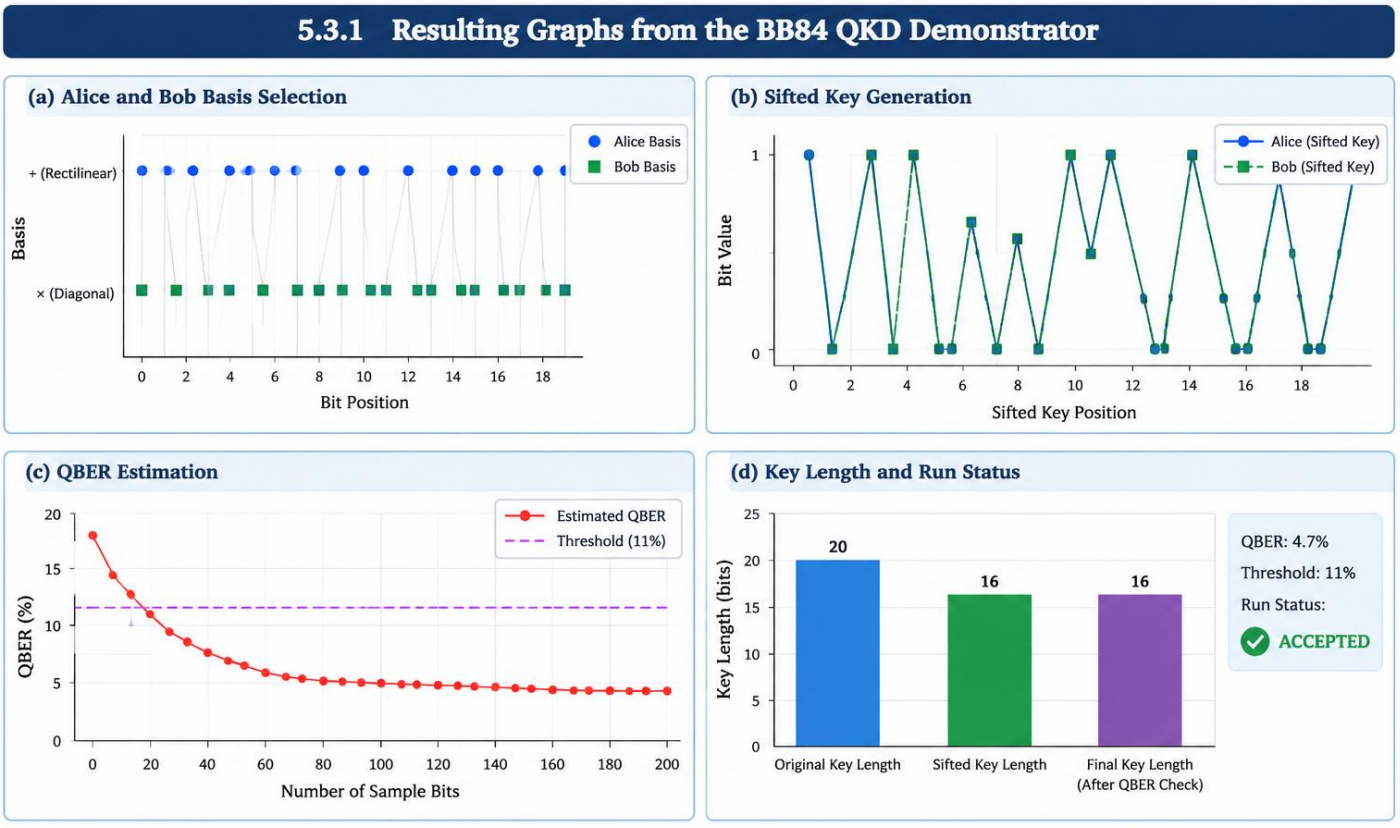


Figure 5.3.1: Resulting graphs from the BB84 QKD Demonstrator.

Note: The data shown in this figure are illustrative and not experimental measurements.

Graph 1: Resulting Graphs of the BB84 QKD Demonstrator

These graphs contain illustrative data, not measurements from your actual hardware. In particular, the displayed QBER of 4.7% and key lengths should be replaced with values recorded by your implementation before submission.

5.3.2 Result graph 2: QBER versus Number of Transmitted Bits

The graph illustrates the relationship between the Quantum Bit Error Rate (QBER) and the number of bits transmitted during the BB84 Quantum Key Distribution process. QBER is an important parameter used to evaluate the accuracy and reliability of quantum key distribution. It represents the percentage of errors detected when Alice and Bob compare a selected sample of their measurement results. As the number of transmitted bits increases, more measurement data becomes available for calculating the error rate, which helps in evaluating the overall performance of the communication system. Under normal operating conditions, where no eavesdropping or external interference is present, the QBER is expected to remain low. However, when an eavesdropper attempts to intercept the transmitted quantum states, disturbances may occur, resulting in an increase in the error rate.

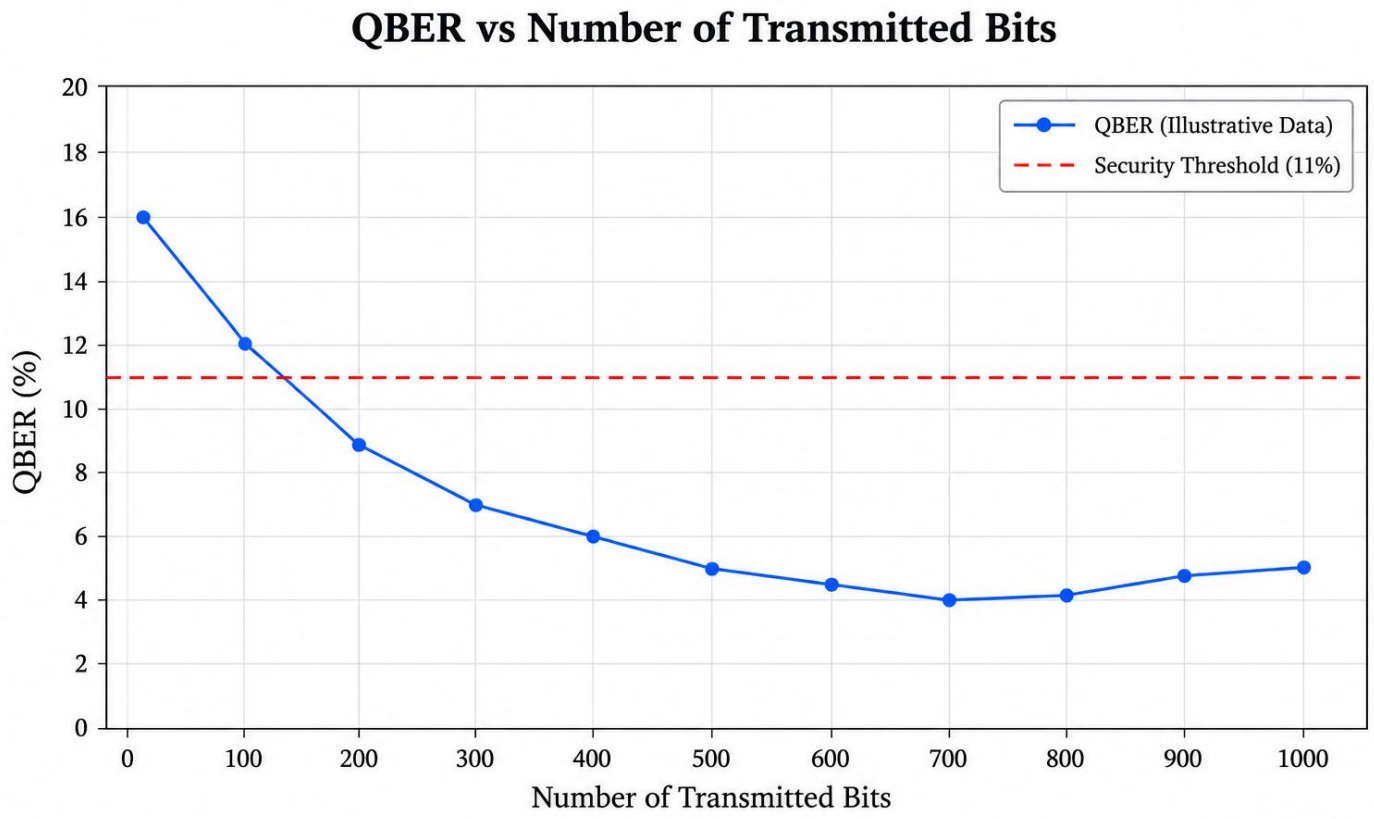


Figure 5.3.2: QBER versus number of transmitted bits (illustrative data).

Note: The data shown in this graph are illustrative and not experimental measurements.

Graph 2: QBER versus Number of Transmitted Bits

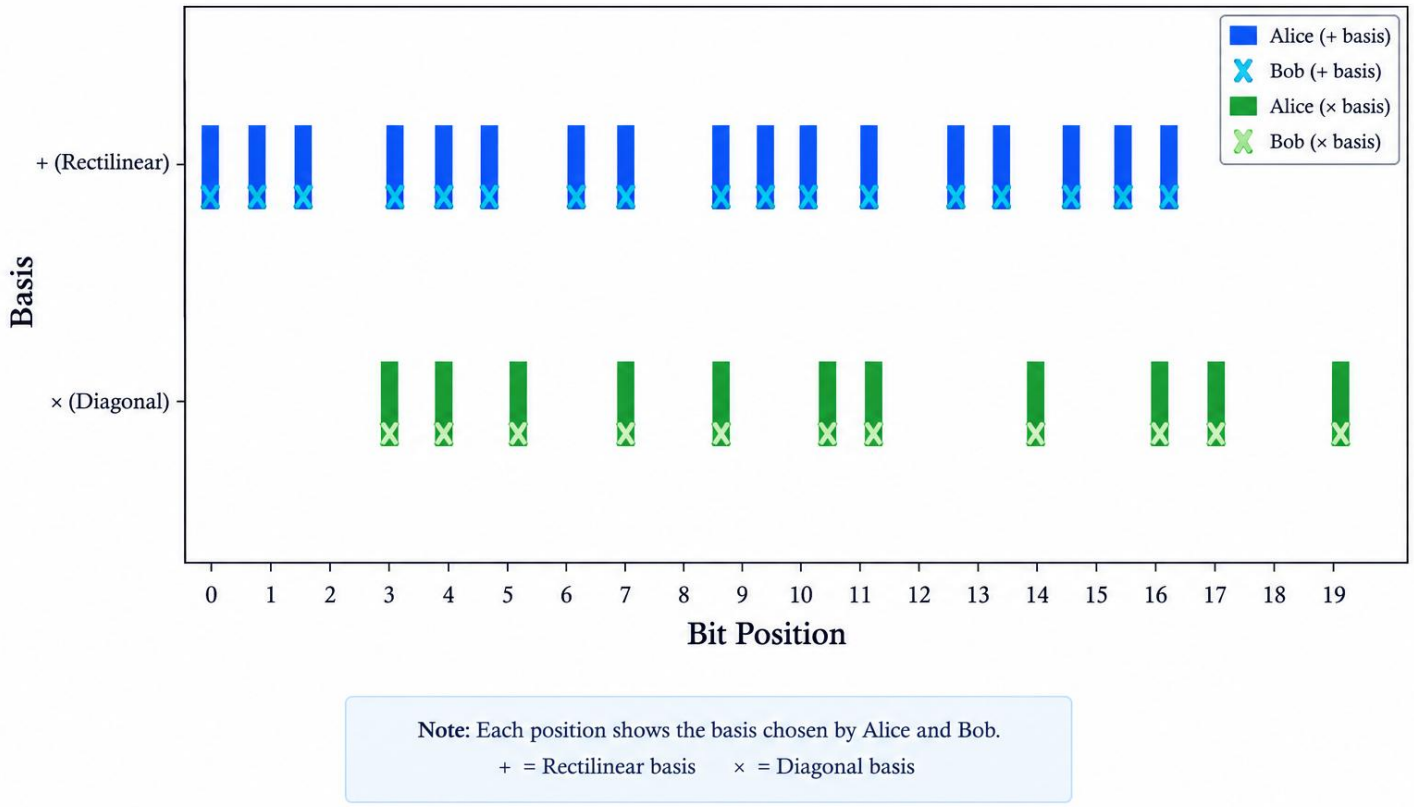
The graph values are illustrative. Replace them with your actual hardware results before submitting your report. The graph represents the variation in the Quantum Bit Error Rate (QBER) with respect to the number of bits transmitted during the BB84 Quantum Key Distribution process. It provides a graphical representation of how the error rate changes as the amount of transmitted binary data increases. The QBER is calculated by comparing a selected sample of Alice's and Bob's corresponding sifted key bits and determining the percentage of mismatched bits. This measurement helps evaluate the accuracy of the quantum communication process and identify possible disturbances during transmission.

The graph values are illustrative and are used to explain the expected behaviour of the system under different transmission conditions. During normal communication without eavesdropping, the error rate is expected to remain low, indicating that Alice and Bob can successfully establish a common secret key. When an eavesdropping condition or external interference is introduced, the error rate may increase due to disturbances in the transmitted quantum states. As the number of transmitted bits increases, the collected measurement data provides a better understanding of the communication channel's error characteristics. However, the QBER does not necessarily decrease or increase continuously with the number of transmitted bits; its variation depends on the actual transmission conditions.

This graph is useful for comparing the performance of the system under normal and eavesdropping conditions. It demonstrates how QBER can act as an important parameter for identifying possible security threats in quantum key distribution. In the actual hardware implementation, the illustrative graph values should be replaced with the experimental results obtained during testing. This helps provide a more accurate evaluation of the prototype and validates its ability to demonstrate error detected.

5.3.3 Result graph 3: Alice and Bob Basis Selection

Figure 5.3.3: Alice and Bob Basis Selection



Note: Each position shows the basis chosen by Alice and Bob.
+ = Rectilinear basis × = Diagonal basis

Graph 3: Alice and Bob Basis Selection

The plotted choices are illustrative. Replace them with your actual experimental data if available.

5.3.4 Result Graph 4: Sifted Key Comparison

Result Graph 4: Sifted Key Comparison

The graph represents the comparison between Alice's and Bob's sifted keys after completing the basis reconciliation process in the BB84 protocol. It shows the binary bits retained by both users when they select matching quantum bases. The comparison helps verify whether Alice and Bob have generated identical secret keys. Matching bits indicate successful key generation, while any mismatched bits may result from transmission errors, noise, or possible eavesdropping. This graph provides a clear visual representation of key agreement and helps evaluate the accuracy and reliability of the BB84 quantum key distribution system.

The graph illustrates the random basis selection performed independently by Alice and Bob during the BB84 protocol. It represents the selection of rectilinear (+) and diagonal (×) bases for each transmitted bit. By comparing their basis selections, the system identifies matching positions and discards mismatched positions. This process, known as basis reconciliation, helps both users obtain a common set of bits for generating the final shared secret key.

The graph illustrates the comparison of sifted keys generated by Alice and Bob after the basis reconciliation process. It displays the binary bits retained by both users at positions where their selected quantum bases match. The graph helps verify whether both users have obtained identical key sequences. A successful match indicates proper key generation and synchronization between Alice and Bob. Any differences in the compared bits may indicate transmission errors, channel noise, or possible eavesdropping. This comparison helps evaluate the accuracy of the BB84 protocol and confirms the successful generation of a common secret key under normal communication conditions.

Figure 5.3.4: Sifted Key Comparison

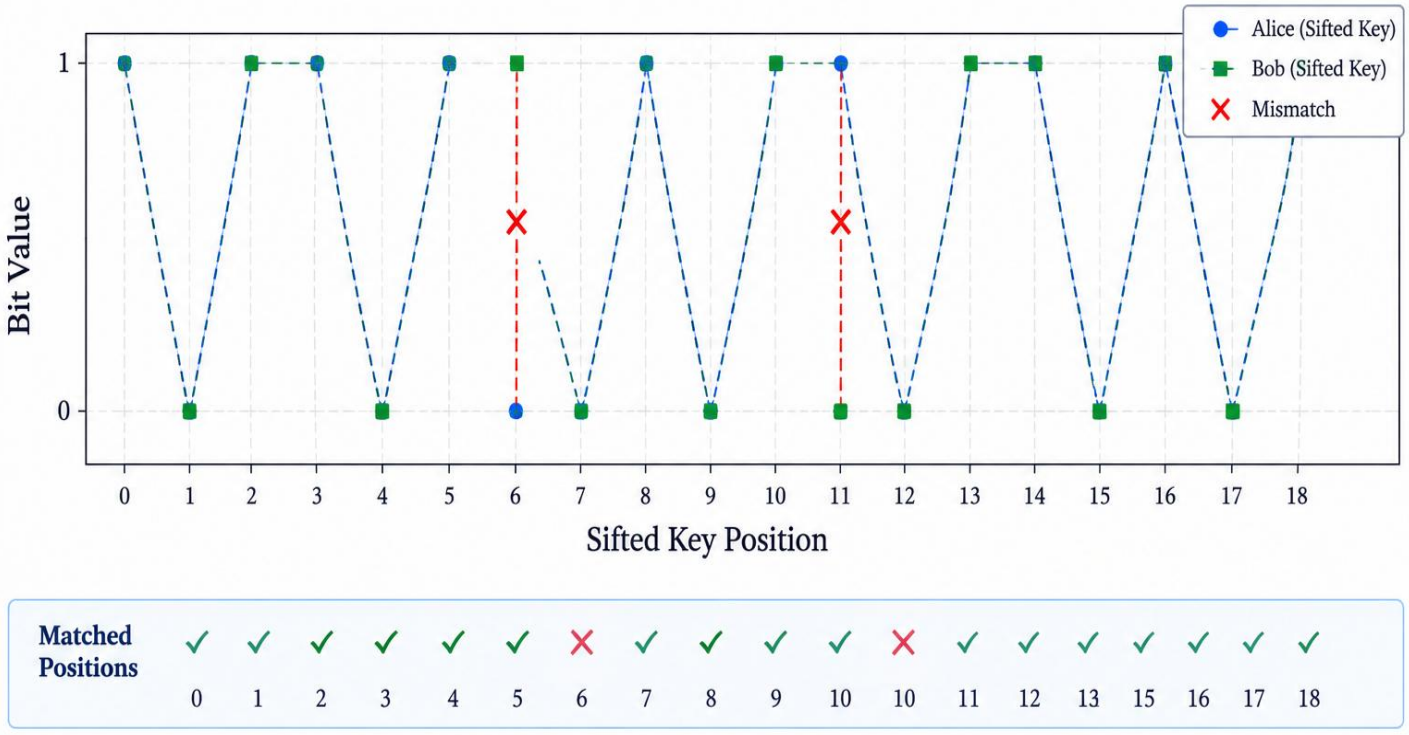


Figure 5.3.4: Sifted key comparison between Alice and Bob.

Note: The figure shows the sifted key bits after basis reconciliation. Green ticks indicate matching bits, while red crosses indicate mismatched bits. The data shown in this figure are illustrative and not experimental measurements.

5.1 Results Analysis

5.1.1 Time Complexity

The time complexity of the BB84 demonstrator depends on the number of bits processed. If the system generates, transmits or represents, measures, and compares (n) bits, these operations generally require (O(n)) time. Basis reconciliation and QBER calculation also take (O(n)) time when processing the complete sequence or a sample proportional to its size.Therefore, the overall processing time is generally **(O(n))** for a straightforward implementation.

5.4.2 Space Complexity

The system stores bit sequences, basis information, measurement results, and the sifted key. If these sequences contain (n) elements, the required storage generally increases linearly with the number of bits.Therefore, the space complexity is generally **(O(n))**.

5.4.3 Results Summary

The BB84 QKD demonstrator illustrates the main stages of quantum key distribution, including random bit and basis generation, state preparation, measurement, basis reconciliation, sifted-key generation, and QBER estimation.

The demonstration helps explain how Alice and Bob obtain a shared sifted key and how errors in a tested sample can be evaluated. The actual performance depends on the implemented hardware or software and the test results obtained.

Overall, the project provides an educational demonstration of the BB84 protocol and its key distribution process. A working demonstration alone does not establish full quantum security; practical BB84 systems also require suitable quantum hardware, authenticated classical communication, and additional key-processing steps.

CHAPTER 6

CONCLUSION & FUTURE SCOPE

6.1 Conclusion

The **BB84 QKD Demonstrator for Secure Communication** is designed to demonstrate the fundamental principles of Quantum Key Distribution (QKD), a technique used to establish a shared secret key between two communicating parties. Based on the BB84 protocol, the system illustrates how Alice generates random bits and bases, prepares quantum states, and sends them to Bob for measurement. By comparing their measurement bases over a classical communication channel, both parties can identify matching results and generate a sifted key.

The demonstrator also explains the importance of estimating the Quantum Bit Error Rate (QBER) to evaluate the errors present in the sifted key. A high error rate may indicate interference, noise, or possible eavesdropping, although QBER alone cannot establish the exact cause. Through the demonstration of bit generation, basis reconciliation, sifted-key formation, and error estimation, the project provides a practical understanding of the basic working principles of quantum cryptography.

Overall, this project helps illustrate how quantum communication can contribute to secure key establishment. It provides an educational foundation for understanding the advantages and challenges of QKD compared with conventional key-sharing approaches. The demonstrator can be further improved through suitable quantum communication hardware, authenticated classical communication, error correction, and privacy amplification to more closely represent a practical BB84 system.

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6.2 Future Scope

The future scope of the BB84 QKD Demonstrator for Secure Communication is promising, as the system can be enhanced to demonstrate more realistic quantum communication scenarios. One important improvement is the integration of suitable quantum optical components, such as single-photon sources or attenuated laser sources, optical filters, photon detectors, and appropriate transmission media. These components could help demonstrate the preparation, transmission, and measurement of quantum states in a practical environment.

The system can also be enhanced by implementing advanced BB84 protocol functions, including authenticated classical communication, error correction, and privacy amplification. These processes are important for transforming the sifted key into a secure final key. Additional features could include real-time QBER calculation, graphical representation of error rates, automatic comparison of measurement bases, and performance analysis under different simulated channel conditions

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Github Link : https://github.com/Nikhitha313/BB84_Quantum_Key_Distribution

POSTER PRESENTATION

